NONSOFIC GROUPS EXIST

ANONYMOUS

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Abstract. We prove that nonsofic groups exist. The group $\mathrm{EL}_4(L_{\mathbb{F}_2}(1,2))$ of elementary matrices over the binary Leavitt algebra is infinite, finitely generated, has Kazhdan's property (T), and is not sofic; the proof is engineered for verification: the two auxiliary matrices on which the construction turns are written as explicit products of elementary matrices, all remaining memberships reduce to one Whitehead identity and the perfectness of Thompson's group V, and the headline result is independent of K-theory.

The analytic engine is a compression–centralizer criterion. If a Kazhdan group is generated by an infinite Kazhdan subgroup Γ together with finitely many elements that conjugate Γ into itself, then in a hypothetical sofic approximation the expander component decompositions supplied by Kun's theorem for Γ and for the ambient group can be matched almost bijectively. The matching is conservative: a median normalization of component sizes is almost monotone along compressor arcs, exact conservation under permutations makes the estimate two-sided, and an elementary ℓ^1 co-area inequality on the ambient expanders pins the normalized sizes at their median. Localizing to a single matched component yields a sofic approximation of $K \times J$—K the compressed image of Γ, J a commuting subgroup—whose K-edges form an essential expander sequence, and the Kun–Thom centralizer obstruction forces J to be LEF. The Leavitt relations realize the scheme with J a copy of Thompson's group V, which is infinite, simple, and finitely presented, hence not LEF.

Matrix self-similarity of $L = L_k(1,2)$ spreads the conclusion: an explicit rank equivalence identifies the elementary groups of all ranks ≥ 2, so the theorem above holds for every $\mathrm{EL}_{m+1}(L_{\mathbb{F}_2}(1,2))$, $m \geq 1$; and for every finite field k, the unit group $L^\times$, every $\mathrm{GL}_r(L)$ with $r \geq 1$, and every $\mathrm{EL}_r(L)$ with $r \geq 2$ are mutually isomorphic infinite finitely generated property (T) nonsofic groups, with the construction already closing in rank two. A finite multiplication-table construction then shows that every finitely generated nonsofic group is a quotient of an infinite finitely presented nonsofic group, Kazhdan whenever the original group is. In particular, not every finitely presented discrete group is sofic.

An appendix gives an elementary and effective proof, apparently the first, that $K_1(L_k(1,2)) = 0$ over every field. A second appendix describes the accompanying Lean 4 formalization, which verifies the principal results and, going further, proves internally every theorem this paper cites—including the decomposition theorems for sofic approximations of Kazhdan groups, property (T) for the elementary groups used, and Shalom's finite-presentation theorem—after repairing two gaps found in the published proofs.

Contents

1. Introduction

A countable discrete group is *sofic* if every finite portion of its multiplication table admits arbitrarily accurate and asymptotically faithful models by permutations of finite sets. The notion arose in Gromov's work on symbolic dynamics [10]; Weiss introduced the term and asked whether every countable group is sofic [21]. The class contains all amenable and all residually finite groups, and it is stable under a long list of standard constructions; see [7, 18]. Whether every group is sofic has remained a central open problem of modern group theory; for a recent account of its status, see [8].

We answer the question in the negative.

Main results. Let k be a field and let

$$L = L_k(1,2) = k\langle s_0, s_1, t_0, t_1 \mid t_i s_j = \delta_{ij},\ s_0t_0 + s_1t_1 = 1\rangle \tag{1}$$

be the binary Leavitt algebra [15]. For a unital ring A and $r \geq 2$, write

$$x_{ij}(a) = I_r + aE_{ij} \in \mathrm{GL}_r(A), \qquad a \in A, \quad i \neq j,$$

and let $\mathrm{EL}_r(A) \leq \mathrm{GL}_r(A)$ be the subgroup generated by these elementary matrices.

MainResults
`universalLeavittEL4_not_isSofic`
`universalLeavitt_profile`
`ambient_profile`

Theorem A. *Let $R = L_{\mathbb{F}_2}(1,2)$. For every $m \geq 1$, the group*

$$\mathrm{EL}_{m+1}(R)$$

is infinite, finitely generated, has property (T), and is not sofic. In particular $G = \mathrm{EL}_4(R)$ is not sofic, and not every countable discrete group is sofic.

The proof of Theorem A is engineered for audit. The two auxiliary matrices of the construction are exhibited as explicit products of elementary matrices; every remaining elementary membership follows from one explicit Whitehead identity together with the perfectness of Thompson's group V; the ranks below four are reached by an explicit isomorphism between the elementary groups of any two ranks ≥ 2 over R (Proposition 5.8); and no K-theory is used.

MainResults
`binaryLeavitt_finiteField_profile`
`binaryLeavittUnits_not_isSofic`
`binaryLeavittGL_not_isSofic`

Theorem B. *Let k be a finite field and let $L = L_k(1,2)$.*

(i) *The unit group $L^\times$ is infinite, finitely generated, has property (T), and is not sofic.*

(ii) *For every $r \geq 1$, every ordered complete binary leaf set $\mathcal{C} = (\alpha_1, \dots, \alpha_r)$ induces an explicit group isomorphism*

$$\Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \xrightarrow{\cong} L^{\times}, \qquad (a_{ij}) \longmapsto \sum_{i,j=1}^{r} \alpha_i a_{ij} \alpha_j^{*}. \tag{2}$$

For $r \geq 2$ one has $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$. Hence all the groups $\mathrm{GL}_r(L)$, $r \geq 1$, and $\mathrm{EL}_r(L)$, $r \geq 2$, are mutually isomorphic, and each is an infinite finitely generated property (T) nonsofic group.

Theorem C. *Every finitely generated nonsofic group is a quotient of an infinite finitely presented nonsofic group. Every nonsofic group with property (T) is a quotient of an infinite finitely presented nonsofic group with property (T). In particular, there is an infinite finitely presented property (T) group H that is not sofic; H may be chosen to surject onto $L^{\times}$, and hence to have a quotient isomorphic to every group listed in Theorem B.*

MainResults
`exists_finitelyPresented_nonsofic_group`
`exists_infinite_finitelyPresented_nonsofic_ambient_cover`
KazhdanCover
`exists_kazhdan_finitelyPresented_cover_of_not_isSofic`

Strategy. The proof has three independent layers.

The first layer is analytic.

Theorem D (Kazhdan compression–centralizer criterion)**.** *Let G be a finitely generated property-(T) group, and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup. Suppose there is a finite set $Q \subseteq G$ with*

$$G = \langle \Gamma, Q \rangle \qquad \text{and} \qquad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q).$$

Fix $q_0 \in Q$, put $K = q_0 \Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$[K, J] = 1, \qquad K \cap J = \{1\}.$$

If G is sofic, then J is LEF.

CriterionAssembly
`isLEF_of_isSofic`
`not_isSofic_of_not_isLEF`

The point of departure is the tension between the two theorems that govern sofic approximations of Kazhdan groups. Kun's theorem [13] proves Bowen's conjecture: every sofic approximation of an infinite property-(T) group is essentially a vertex-disjoint union of uniformly expanding components. The Kun–Thom theorem [14] requires more: if the restriction of a sofic approximation of $\Lambda \times \Delta$ to a Kazhdan factor Λ is essentially *one* expander, then Δ is locally embeddable into finite groups (LEF). A disjoint union of equal-sized expanders is not enough, because the commuting factor may permute the components—this is exactly what happens in a product approximation.

Compression closes the gap. A hypothetical sofic approximation of G yields a Kun decomposition for Γ and another for G. Transporting the Γ-components by a compressor q gives one-sided inclusions into other Γ-components. Write $s(y)$ for the size of the Γ-component containing y; on each ambient G-component, normalize by a vertex-weighted median m of the sizes and set

$$f(y) = \frac{s(y)}{s(y) + m}. \tag{3}$$

The one-sided inclusions make f almost monotone along compressor arcs. But a compressor acts by a permutation, so total increase equals total decrease *exactly*. The resulting small ℓ^1 edge variation, fed into an elementary co-area inequality on the ambient expanders, pins f at its median value $1/2$. Hence transported and target components have asymptotically equal sizes, and one-sided containment becomes almost-bijective matching. A weighted diagonal selection extracts a single matched component on which every multiplication, faithfulness, invariance, conjugacy, and graph-edit error is negligible. Restricting and completing the permutations yields a sofic approximation of $K \times J$ whose K-edges form an essential expander sequence, and the Kun–Thom obstruction forces J to be LEF. Theorem D is proved in Section 4 as a corollary of a local statement, Theorem 4.1, which isolates the exact graph-theoretic hypotheses and records precisely what property (T) is used for. Remarks 4.6 and 4.7 test the criterion against sofic examples and show that each hypothesis earns its place.

The second layer is algebraic, and it is where the Leavitt relations enter. Fix $m \geq 2$ and a nontrivial unital ring A carrying a binary Leavitt family—four elements satisfying the two Leavitt relations; no copy of L is assumed. A left-comb binary partition with $m+1$ leaves produces a compressor $u \in \mathrm{GL}_{m+1}(A)$ that conjugates the block subgroup $\Gamma \cong \mathrm{EL}_m(A)$ into the corner $a \mapsto s_0 a t_0$ of itself. A block involution z centralizes the compressed image K, so $v = zu$ is a second compressor with the same image, and $vu^{-1} = z$ generates every elementary matrix involving the last coordinate; whenever u and z lie in $\mathrm{EL}_{m+1}(A)$, it follows that $G = \mathrm{EL}_{m+1}(A) = \langle \Gamma, u, v \rangle$. The complementary corner $a \mapsto p_0 + s_1 a t_1$ carries a commuting copy of Thompson's group V, which is infinite, simple, and finitely presented, hence not LEF. The corner copy of V is elementary over every ambient ring: V is perfect, and the Whitehead lemma places diagonal products of commutators of units in EL_2. The only membership hypotheses of the general theorem therefore concern u and z. Over $\mathbb{F}_2$ they are discharged by explicit four-term elementary factorizations, and an explicit rank equivalence between the elementary groups of any two ranks ≥ 2—block flattening along the self-similarity of the Leavitt relations—then spreads the conclusion from rank four to every rank, which gives Theorem A with no K-theory. Over every finite field the memberships also follow from $K_1(L) = 0$ and the GE property of purely infinite simple rings; the K_1 input itself admits an elementary, effective proof, written out in full in Appendix A. Matrix self-similarity $M_r(L) \cong L$ then spreads nonsoficity and property (T) to the unit group and to all GL_r, and a separate two-by-two realization shows that the scheme already closes in rank two.

The third layer is a finite multiplication-table construction. Every finitely generated nonsofic group G has a finite table fragment that admits no sufficiently accurate permutation model. Imposing precisely that finite table as relators gives a finitely presented group mapping onto G; the map proves infinitude and preserves distinctness of the named elements, while the forbidden finite table proves nonsoficity directly. When G is Kazhdan, Shalom's finite-presentation theorem first supplies a finitely presented Kazhdan cover, and imposing the same table inside its kernel produces a finitely presented Kazhdan nonsofic cover. The covers are proved nonsofic directly, from the forbidden table itself; quotient closure of soficity, which is unknown, plays no role.

External inputs, and how to verify this paper. The proof is organized so that its trust surface is explicit and so that a skeptical reader can audit it in layers.

A machine-checked audit is available as a second, independent layer, and it is reported in the margins: each numbered result below carries a note naming the modules and declarations of the accompanying Lean development that formalize it, or recording that it is formalized only in part, or that it is not formalized and why. The notes are marginal in the strict sense—no argument in this paper appeals to one, and deleting all of them leaves the text unchanged—so a reader uninterested in formalization can ignore them throughout. Appendix B says what the development proves, what it assumes, and where it is stronger than the text.

On the paper layer, Theorems A and D rest on the following external inputs: Kun's expander decomposition theorem [13], cited in one-way form and with two proof repairs recorded in Remark 2.11; the Kun–Thom centralizer obstruction [14], with the repair recorded there as well; property (T) for elementary groups over finitely generated rings [9]; and classical facts about Thompson's group V—infinite, simple, finitely presented, and 2-generated [11, 6, 5]. Everything else is self-contained. In the proof of Theorem A, the two auxiliary matrices u and z are factored explicitly into elementary matrices, and every remaining elementary membership follows from the explicit Whitehead identity of Lemma 5.5 together with the perfectness of V. The analytic sections use no spectral theory: the only analytic inequality proved and used in the matching argument is the ℓ^1 co-area estimate of Lemma 2.7, established in a few lines from the layer-cake formula, while the approximation-theoretic depth is confined to the cited theorems of Kun and Kun–Thom.

On the machine layer, *none of these inputs is external.* The accompanying Lean development proves internally, in the exact forms used here: the one-way Kun decomposition, with both repairs; the Kun–Thom obstruction, in the essential form of Theorem 2.10; property (T) for the rank-three elementary groups it uses, over every finite field, with an explicit Kazhdan pair; the non-LEF property of the witness, by a direct finite obstruction that bypasses simplicity and finite presentability of V entirely (Remark 5.13); and Shalom's finite-presentation theorem. The endpoints of the development are closed theorems whose axioms are the three axioms of classical mathematics in Lean. So the citations above carry this paper's proofs and the credit for the underlying mathematics, while the formalization independently re-proves each cited input; a reader may weight the two layers as they see fit.

Theorem B additionally uses $K_1(L) = 0$ and the GE property of purely infinite simple rings [1, 3, 2]. Neither remains a dependency: the GE property is proved at every rank from strong division (Lemma A.4), and Appendix A proves $K_1(L) = 0$ elementarily and effectively, so the K-theoretic citations are retained as independent confirmation, and [3] is invoked as a dependency only for the perfectness of $L^\times$ (Proposition 8.3(ii)), which no theorem of this paper consumes.

Theorem C is self-contained apart from Shalom's finitely presented Kazhdan covers [19], used only for the Kazhdan refinement; the formalization proves Shalom's theorem internally as well.

A reader who wants to check the headline claim should verify, in order: the five finite lemmas of Section 3, each elementary and a few lines long; the proof of Theorem 4.1 in Section 4, which assembles them and invokes Kun's theorem twice and the Kun–Thom theorem once, in exactly the forms stated in Section 2.5; the matrix identities of Sections 5 and 6, each a finite computation from the two Leavitt relations; and the two explicit factorizations of Section 7. That path proves Theorem A in full. Sections 8 and 9 lie downstream and can be audited separately.

Organization. Section 2 fixes the sofic, expander, and LEF vocabulary and states the theorems of Kun and Kun–Thom in the forms used. Section 3 isolates the five elementary lemmas of the matching mechanism. Section 4 proves the local compression–centralizer theorem and deduces Theorem D. Section 5 develops Leavitt leaf calculus, matrix self-similarity, the elementary rank equivalence, the Whitehead lemma, the characteristic-two block factorizations, and the corner realization of V. Section 6 presents the compression scheme in adjacent ranks over an arbitrary ring carrying a binary Leavitt family and proves the Leavitt-corner theorem. Section 7 discharges all memberships explicitly over $\mathbb{F}_2$ and proves Theorem A. Section 8 proves the two-by-two theorem, the equality $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$, and Theorem B. Section 9 proves Theorem C. Section 10 collects concluding remarks. Appendix A proves $K_1(L_k(1, 2)) = 0$ elementarily and effectively; Appendix B documents the formalization.

2. Sofic approximations, expanders, and LEF groups

2.1. **Sofic approximations.** For $\sigma, \tau \in \mathrm{Sym}(Y)$, let

$$d_{\mathrm{H}}(\sigma, \tau) = \frac{1}{|Y|} \, |\{y \in Y \mid \sigma(y) \neq \tau(y)\}|$$

denote the normalized Hamming distance.

Sofic
`SoficApproximation`
`IsSofic`

Definition 2.1. Let G be a countable discrete group. A *sofic approximation* of G is a sequence of maps $\sigma_n : G \to \mathrm{Sym}(Y_n)$, with $|Y_n| \to \infty$, such that

(a) $d_{\mathrm{H}}(\sigma_n(g)\sigma_n(h), \sigma_n(gh)) \to 0$ for all $g, h \in G$;

(b) $d_{\mathrm{H}}(\sigma_n(g), \mathrm{id}) \to 1$ for all $g \in G \setminus \{1\}$.

The group G is *sofic* if it admits a sofic approximation.

The maps σ_n are not assumed to be homomorphisms. Property (b) self-improves: for a sofic group one may always pass to a sofic approximation in which each $\sigma_n(g)$, $g \neq 1$, moves *every* point of a set of density $1 - o(1)$, and it is standard that (a) and (b) force $d_{\mathrm{H}}(\sigma_n(1), \mathrm{id}) \to 0$. We will use the following exact normalization.

Lemma 2.2 (Finite normalization)**.** *Let $\sigma_n : G \to \mathrm{Sym}(Y_n)$ be a sofic approximation and let $F \subseteq G$ be a finite symmetric set. Then there is a sofic approximation $\sigma'_n : G \to \mathrm{Sym}(Y_n)$ with $d_{\mathrm{H}}(\sigma_n, \sigma'_n) \to 0$ pointwise, such that for all n:* Sofic `map_one_close` *InverseNormalization* `inverseError_negligible`

$$\sigma'_n(1) = \mathrm{id}, \qquad \sigma'_n(g^{-1}) = \sigma'_n(g)^{-1} \quad (g \in F).$$

Proof. Set $\sigma'_n(1) = \mathrm{id}$, and note that $d_{\mathrm{H}}(\sigma_n(1), \mathrm{id}) \to 0$: applying (a) with $g = h = 1$ gives $d_{\mathrm{H}}(\sigma_n(1)^2, \sigma_n(1)) \to 0$, and composing with the permutation $\sigma_n(1)^{-1}$ preserves d_{H}.

Choose one representative from each pair $\{g, g^{-1}\} \subseteq F \setminus \{1\}$ with $g \neq g^{-1}$; put $\sigma'_n(g) = \sigma_n(g)$ and $\sigma'_n(g^{-1}) = \sigma_n(g)^{-1}$. Then $d_{\mathrm{H}}(\sigma_n(g^{-1}), \sigma_n(g)^{-1}) \to 0$, because $\sigma_n(g^{-1})\sigma_n(g)$ is d_{H}-close to $\sigma_n(1)$, hence to id, and d_{H} is invariant under composition with the permutation $\sigma_n(g)^{-1}$ on the right.

If $g = g^{-1} \in F \setminus \{1\}$ is an involution, let $A_n = \{y \mid \sigma_n(g)^2 y = y\}$. Since $\sigma_n(g)^2$ is d_{H}-close to $\sigma_n(g^2) = \sigma_n(1)$, hence to id, we have $|Y_n \setminus A_n| = o(|Y_n|)$. The set A_n is $\sigma_n(g)$-invariant—if $\sigma_n(g)^2 y = y$, then applying $\sigma_n(g)$ gives $\sigma_n(g)^2(\sigma_n(g)y) = \sigma_n(g)y$—and $\sigma_n(g)$ restricts to an involution of A_n. Define $\sigma'_n(g)$ to be $\sigma_n(g)$ on A_n and the identity on $Y_n \setminus A_n$; this is an involution of Y_n differing from $\sigma_n(g)$ on $o(|Y_n|)$ points.

Outside $F \cup \{1\}$ put $\sigma'_n = \sigma_n$. Each $\sigma'_n(g)$ differs from $\sigma_n(g)$ on $o(|Y_n|)$ points, so properties (a) and (b) persist. $\square$

The next lemma is the bridge, used many times below, between statements about products of letter permutations and statements about the permutation of the product.

Lemma 2.3 (Word approximation)**.** *Let $\sigma_n : G \to \mathrm{Sym}(Y_n)$ be a sofic approximation, let $g \in G$, and let $g = t_1 t_2 \cdots t_\ell$ be any fixed factorization with $t_1, \ldots, t_\ell \in G$. Then* Sofic `word_close` *BlockWordCrossing* `wordDisagreement_negligible` `all_wordCrossing_negligible`

$$\big|\{y \in Y_n \mid \sigma_n(t_1)\sigma_n(t_2) \cdots \sigma_n(t_\ell)\, y \neq \sigma_n(g)y\}\big| = o(|Y_n|).$$

Proof. Induction on ℓ, the case $\ell = 1$ being trivial. For $\ell \geq 2$ put $h = t_2 \cdots t_\ell$. Composing with the permutation $\sigma_n(t_1)$ on the left does not change disagreement sets, so $\sigma_n(t_1) \cdots \sigma_n(t_\ell)$ agrees with $\sigma_n(t_1)\sigma_n(h)$ outside the $o(|Y_n|)$ points supplied by the inductive hypothesis; and $\sigma_n(t_1)\sigma_n(h)$ agrees with $\sigma_n(t_1 h) = \sigma_n(g)$ outside $o(|Y_n|)$ points by Definition 2.1(a). The union of the two sets has size $o(|Y_n|)$. $\square$

Whenever an element of a subgroup is written below as a word in a generating set and a property is verified letter by letter, Lemma 2.3 is what transfers the conclusion from the product of the letter permutations to the permutation of the element itself; it is cited explicitly at each such point.

2.2. **Generator graphs and edit distance.** Let $T \subseteq G$ be a finite set and $\sigma : G \to \mathrm{Sym}(Y)$ a map. The *T-generator graph* $X(Y, \sigma, T)$ is the multigraph on the vertex set Y with one edge $\{y, \sigma(t)y\}$ for each pair $(y,t) \in Y \times T$ with $\sigma(t)y \neq y$; parallel edges are retained, fixed points contribute no edge, and every vertex has degree at most $2|T|$. An *arc* of $t \in T$ is an ordered pair $(y, \sigma(t)y)$.

Conventions. All graphs in this paper are finite multigraphs without loops, and all edge counts—degrees, boundaries, edit distances, and sums over $E(X)$—count edges with multiplicity. For multigraphs X, X' on the same vertex set Y, the *edit distance* $\mathrm{ed}(X, X')$ is the minimal number of edge insertions and deletions, with multiplicity, transforming X into X'. A sequence of graphs X_n on Y_n is *edit-equivalent* to a sequence X'_n if $\mathrm{ed}(X_n, X'_n) = o(|Y_n|)$.

For a finite multigraph X and $\emptyset \neq U \subseteq V(X)$, write ∂U for the multiset of edges with exactly one endpoint in U, and

$$h(X) = \min\left\{ \frac{|\partial U|}{|U|} \,\middle|\, \emptyset \neq U \subseteq V(X),\ |U| \leq \tfrac{1}{2}|V(X)| \right\}$$

for the Cheeger constant; for a one-vertex graph the minimum runs over the empty set, and we set $h(X) = +\infty$. A sequence of bounded-degree graphs X_n with $|V(X_n)| \to \infty$ is an *expander sequence* if $\inf_n h(X_n) > 0$.

Remark 2.4 (Robustness of the conventions)**.** The conventions are robust in exactly the two directions in which they interface with the cited external theorems. Loops are immaterial: a loop lies in no boundary ∂U, so inserting or deleting loops changes no Cheeger constant, no component, and no estimate below; arcs with $\sigma(t)y = y$ are accordingly discarded once and for all in the definition above. Multiplicities are controlled by a uniform bound: in a T-generator graph every unordered pair carries at most $2|T|$ parallel edges, one per arc. For multigraphs of multiplicity at most M: forgetting multiplicities preserves components and connectivity; boundary sizes change by a factor between 1 and M, so a componentwise Cheeger bound h in one convention yields at least h/M in the other; and one simple-graph edit lifts to at most M multigraph edits, so $o(|Y_n|)$ edit-distance statements transfer in both directions. We do not assert any comparison for arbitrary multigraphs—two multigraphs can share an underlying simple graph and differ by linearly many parallel copies; the transfer above uses the multiplicity bound, which holds for every graph in this paper. Consequently every statement below, and each of the two external theorems of Section 2.5, may be read in either convention with constants adjusted; we fix the multiplicity-counting convention throughout.

Definition 2.5. Let (X_n) be a sequence of graphs of uniformly bounded degree on vertex sets Y_n with $|Y_n| \to \infty$. An *expander decomposition* of (X_n) consists of graphs $\widetilde{X}_n$ on Y_n, of uniformly bounded degree, with $\mathrm{ed}(X_n, \widetilde{X}_n) = o(|Y_n|)$, together with a constant $h > 0$ such that every connected component of every $\widetilde{X}_n$ has Cheeger constant at least h.

Criterion
`ExpanderDecomposition`

Definition 2.6. A sequence of bounded-degree graphs (X_n) is an *essential expander sequence* if it is edit-equivalent to an expander sequence.

formalized in part
Criterion
`MatchingCertificate`
EssentialExpanderRepair
`induced_expands_eventually`
`repairedActualGraph_expands_eventually`
Realized as the matching certificate expansion data and its repair, not as a separate definition.

2.3. **Expansion and an ℓ^1 co-area estimate.** The single analytic inequality used in this paper is the following co-area estimate.

Lemma 2.7 (ℓ^1 co-area estimate)**.** *Let X be a finite connected graph and $f : V(X) \to \mathbb{R}$. Let $c \in \mathbb{R}$ be such that*

$$|\{x \mid f(x) > c\}| \le \tfrac{1}{2}|V(X)| \qquad \text{and} \qquad |\{x \mid f(x) < c\}| \le \tfrac{1}{2}|V(X)|.$$

Then

$$\sum_{x \in V(X)} |f(x) - c| \ \le\ \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f(x) - f(y)|.$$

FiniteGraph
`coarea`
`coarea_nul`
DirectedCoarea
`nonnegative_coarea`

Proof. If X has a single vertex, the median hypotheses force $f \equiv c$ and both sides vanish (with $1/h(X) = 0$ by convention); assume $|V(X)| \ge 2$. Replacing f by $f - c$ we may assume $c = 0$. Write $f = f_+ - f_-$ with $f_\pm \ge 0$. For $t > 0$, the level set $U_t = \{x \mid f_+(x) > t\}$ satisfies $|U_t| \le \frac{1}{2}|V(X)|$, so $|\partial U_t| \ge h(X)|U_t|$ whenever $U_t \ne \emptyset$. By the layer-cake formula,

$$\sum_x f_+(x) = \int_0^\infty |U_t| \,\mathrm{d}t \le \frac{1}{h(X)} \int_0^\infty |\partial U_t| \,\mathrm{d}t \le \frac{1}{h(X)} \sum_{\{x,y\} \in E(X)} |f_+(x) - f_+(y)|,$$

since an edge $\{x, y\}$ lies in ∂U_t exactly for t in an interval of length $|f_+(x) - f_+(y)|$. The same bound holds for f_-, and $|f_+(x) - f_+(y)| + |f_-(x) - f_-(y)| \le |f(x) - f(y)|$ pointwise on edges, because f_+ and f_- have disjoint supports. Adding the two estimates gives the claim. $\square$

2.4. **LEF groups.** A group J is *LEF* (locally embeddable into finite groups) [20] if for every finite subset $F \subseteq J$ there are a finite group P and a map $\varphi : F \to P$ that is injective and satisfies $\varphi(g)\varphi(h) = \varphi(gh)$ whenever $g, h, gh \in F$. Every LEF group is sofic; finite and residually finite groups are LEF.

Lemma 2.8. *A finitely presented LEF group is residually finite.*

not formalized
Not needed: the non-LEF witness is a direct finite obstruction, not a residual-finiteness argument.

Proof. Let $J = \langle T \mid W \rangle$ with T, W finite and T symmetric, and let $g \in J \setminus \{1\}$; fix a word w_g in T representing g. Let $F \subseteq J$ be the finite set consisting of 1, the elements of T, all partial products $t_1 \cdots t_i$ of every relator $t_1 \cdots t_\ell \in W$, and all partial products of w_g; in particular $g \in F$. Let $\varphi : F \to P$ be a local embedding into a finite group. First, $\varphi(1)\varphi(1) = \varphi(1)$ gives $\varphi(1) = 1_P$. Next, the assignment $t \mapsto \varphi(t)$ kills every relator: its partial products $p_i = t_1 \cdots t_i$ lie in F along with the letters, so $\varphi(p_{i+1}) = \varphi(p_i)\varphi(t_{i+1})$ by multiplicativity, and induction gives $\varphi(t_1)\cdots\varphi(t_\ell) = \varphi(p_\ell) = \varphi(1) = 1_P$. Hence $t \mapsto \varphi(t)$ extends to a homomorphism $\psi : J \to P$. The same induction along the partial products of w_g, all of which lie in F, gives $\psi(g) = \varphi(g)$, and $\varphi(g) \neq \varphi(1) = 1_P$ by injectivity of φ. Thus g survives in a finite quotient, and J is residually finite. □

2.5. **The theorems of Kun and Kun–Thom.** We use two structure theorems for sofic approximations of Kazhdan groups.

KunDecomposition
`exists_expanderDecomposition`
KunFixedDecomposition
`expanderDecomposition`
Proved internally, in the full-sequence one-way form used here; not assumed.

Theorem 2.9 (Kun [13])**.** *Let Γ be an infinite finitely generated property-(T) group with a finite symmetric generating set S, and let $\sigma_n : \Gamma \to \mathrm{Sym}(Y_n)$ be a sofic approximation. Then the S-generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition. Moreover, the modified graphs $\widetilde{X}_n$ may be chosen of degree at most d, with the Cheeger constants of all components bounded below by a single constant $h > 0$; the constants d and h depend only on (Γ, S).*

Kun's theorem proves Bowen's conjecture: sofic approximations of infinite Kazhdan groups are essentially disjoint unions of expanders. Theorem 2.9 is deliberately a *one-way* statement: the four-way equivalence of [13, Theorem 3] contains a spectral implication that fails as written and is not imported anywhere in this paper; Remark 2.11 records this precisely, together with two local repairs to the published proofs of the theorems of this subsection. The companion result of Kun and Thom is an obstruction for centralizers.

KunThomTheorem
`isLEF_of_exactProductExpansion`
KunThomEssential
`isLEF_of_matchingCertificate`
Proved internally, with the matching-core repair; not assumed.

Theorem 2.10 (Kun–Thom [14])**.** *Let Λ be an infinite finitely generated property-(T) group with finite symmetric generating set S_Λ, and let Δ be a finitely generated group. Suppose $\Lambda \times \Delta$ admits a sofic approximation $\pi_n : \Lambda \times \Delta \to \mathrm{Sym}(D_n)$ such that the S_Λ-generator graphs $X(D_n, \pi_n|_\Lambda, S_\Lambda)$ form an essential expander sequence. Then Δ is LEF.*

Theorem 2.10 states the theorem of [14] in its *essential* form, which is exactly how it is invoked in Section 4. The statement in [14] assumes that the Λ-graphs form a genuine expander sequence; the essential form is nonetheless the published theorem's, on two independent grounds. First, the published proof opens under the weaker hypothesis that the sequence is essentially expanding and is carried out under that hypothesis throughout. Second, and independently of any reading of the cited proof, the essential form is proved outright in the accompanying formalization (Appendix B); so the interface used here rests on no charitable interpretation. The heuristic content: a genuine expander is essentially connected, so the commuting factor Δ cannot shuffle components and is forced to act almost by automorphisms of finite structures, which yields LEF approximations.

Interface. The proof in [14] is carried out for *regularly labeled* graphs: each generator label has exactly one outgoing directed edge at every vertex. An approximation normalized as in Lemma 2.2 supplies this structure. Every generator acts by a genuine permutation, and an inverse pair of generators acts by an inverse pair of permutations, so the two directed arcs of a pair combine into one undirected labeled edge; identity loops are discarded by the conventions of Section 2.2. Distinct labels rarely collide: for generators g, h with $h \neq g^{\pm 1}$, a collision $\sigma_n(g)y = \sigma_n(h)y$ makes y a preimage under the permutation $\sigma_n(h)$ of a fixed point of $\sigma_n(h)^{-1}\sigma_n(g)$, which agrees with $\sigma_n(h^{-1}g)$ outside a vanishing fraction of points by approximate multiplicativity and has a vanishing fraction of fixed points by asymptotic faithfulness. These properties are verified for the localized approximation at the end of the proof of Theorem 4.1.

Remark 2.11 (Audit of the two cited proofs)**.** Because the two theorems above carry the entire approximation-theoretic weight of this paper, their published proofs were audited line by line in the course of the formalization, with three findings; we record them here both to delimit exactly what is imported and to credit the arguments of [13, 14], which survive intact.

KunSpectralCounterexample
`switched`
`rayleigh_testVector`
`not_isBipartite`
The near-bipartite counterexample to the spectral implication, formalized.

First, Theorem 3 of [13] is stated as a four-way equivalence, and its spectral implication (4) $\Rightarrow$ (1) fails as written: it invokes the Cheeger–Alon–Milman bound, which separates the nonconstant Markov spectrum from $+1$ but not from -1, and near-bipartite uniform expanders (a two-edge switch in each graph of a bipartite expander family, repeated) satisfy (2)–(4) while violating (1). This does not affect the present paper, nor Theorem 1 of [13]: the decomposition theorem needs only the forward implications, and Theorem 2.9 above is exactly that one-way statement—the only form used here and the form proved in the formalization.

Second, the proof of the forward direction in [13] normalizes a Markov defect (Lemma 10) without first excluding the degenerate case in which that defect vanishes; the repair separates the zero-defect case, where the displacement estimate holds for homogeneous reasons, before dividing.

Third, the centralizer argument of [14] constructs a correspondence between finite models and later treats it as the graph of a permutation, which it need not yet be; the repair measures the row and column fibers of the correspondence, extracts its injective matching core, and completes that core to a genuine permutation, after which the argument proceeds as published.

Both repairs are local to the proofs and are carried out in full in the accompanying formalization (Appendix B), which proves Theorems 2.9 and 2.10 from first principles; a reader auditing this paper by hand should read [13, 14] with the three points above in mind.

3. Conservative component matching: the toolkit

This section isolates five elementary lemmas about finite graphs, permutations, and sequences. They are stated once, in the exact quantitative forms used in Section 4; the only genuinely group-theoretic input of the paper appears there.

3.1. **Expander refinement.**

Refinement
`exists_dominant_cell`
`le_crossing_of_cell`
ComponentRefinement
`refineComponent`

Lemma 3.1 (Expander refinement)**.** *Let X be a finite graph, every connected component of which has Cheeger constant at least $h > 0$. Let $V(X) = A_1 \sqcup \cdots \sqcup A_r$ be a partition, and let e be the number of edges of X whose endpoints lie in different parts. Then there is an assignment $C \mapsto i(C)$ of a part to each component of X, with $|C \cap A_{i(C)}|$ maximal, such that*

$$\sum_C |C \setminus A_{i(C)}| \ \le\ \frac{4e}{h}.$$

Proof. Every edge of X lies inside a component, so the crossing edges distribute among the components; let e_C be the number inside C, with $\sum_C e_C = e$. Fix a component C, let $a_1 \ge a_2 \ge \cdots \ge a_k$ be the sizes of its nonempty intersections with the parts, and let $A_{i(C)}$ realize a_1. We claim

$$|C| - a_1 \ \le\ \frac{4e_C}{h},$$

which proves the lemma upon summation. If C meets only one part, then $C \setminus A_{i(C)} = \emptyset$ and the claim is trivial; assume C meets at least two parts. If a subset $W \subseteq C$ is a union of part-intersections with $|W| \le \frac{1}{2}|C|$, then every edge of $\partial_{X[C]} W$ joins different parts, so $e_C \ge |\partial_{X[C]} W| \ge h|W|$.

If $a_1 \ge \frac{1}{2}|C|$, apply this to $W = C \setminus A_{i(C)}$, a union of part-intersections of size $|C| - a_1 \le \frac{1}{2}|C|$; this gives $|C| - a_1 \le e_C/h$.

If $a_1 < \frac{1}{2}|C|$, accumulate part-intersections in any order until the running total first reaches at least $\frac{1}{4}|C|$; the resulting union W satisfies $\frac{1}{4}|C| \le |W| < \frac{1}{4}|C| + a_1 < \frac{3}{4}|C|$, so $\min(|W|, |C \setminus W|) \ge \frac{1}{4}|C|$, and whichever of W, $C \setminus W$ has size at most $\frac{1}{2}|C|$ is again a union of part-intersections. Applying the boundary bound to it gives $e_C \ge \frac{h}{4}|C| \ge \frac{h}{4}(|C| - a_1)$. □

3.2. **Permutation conservation.**

PermutationConservation
`permutation_conservation_full`

Lemma 3.2 (Permutation conservation)**.** *Let Y be a finite set, $\pi \in \mathrm{Sym}(Y)$, and $f : Y \to \mathbb{R}$. Then*

$$\sum_{y \in Y} \big(f(y) - f(\pi y)\big)_+ \ =\ \sum_{y \in Y} \big(f(\pi y) - f(y)\big)_+ \ =\ \tfrac{1}{2} \sum_{y \in Y} |f(\pi y) - f(y)|.$$

Proof. Both $\sum_y f(y)$ and $\sum_y f(\pi y)$ equal the same number, so $\sum_y (f(y) - f(\pi y)) = 0$; split into positive and negative parts. □

This exact identity is what converts a one-sided (monotonicity) estimate along the arcs of a permutation into a two-sided ℓ^1 estimate.

3.3. **Median pinning.**

Pinning
`median_pinning`
ComponentPinning
`normalized_pinning_global`

Lemma 3.3 (Median pinning)**.** *Let Z be a finite graph in which every connected component B satisfies $h(Z[B]) \ge h > 0$. Let $f : V(Z) \to \mathbb{R}$, and* *suppose that on every component at most half of the vertices satisfy* $f > \frac{1}{2}$ *and at most half satisfy* $f < \frac{1}{2}$*. Then*

$$\sum_{x\in V(Z)} \left|f(x) - \tfrac{1}{2}\right| \;\le\; \frac{1}{h} \sum_{\{x,y\}\in E(Z)} |f(x) - f(y)|.$$

Proof. Every edge of Z lies within a component. Apply Lemma 2.7 with $c = \frac{1}{2}$ on each component and sum. □

3.4. **Weighted selection.**

Lemma 3.4 (Weighted selection)**.** *For each* n*, let* $\{D_{n,i}\}_{i\in I_n}$ *be disjoint nonempty subsets of a finite set* Y_n*,* $|Y_n| = N_n \to \infty$*, with* $\sum_{i\in I_n} |D_{n,i}| \ge (1 - o(1))N_n$*. Suppose that for every fixed* M*,* Selection exists_selection diagonalLevel_error

$$\sum_{i:\,|D_{n,i}|\le M} |D_{n,i}| = o(N_n).$$

Let, for each $r \in \mathbb{N}$*, nonnegative error weights* $E^{(r)}_{n,i}$ *be given with*

$$\sum_{i\in I_n} E^{(r)}_{n,i} = o(N_n) \qquad \textit{for every fixed } r.$$

Then there exist indices $i(n) \in I_n$*, for all* n *in a cofinite set, such that* $|D_{n,i(n)}| \to \infty$ *and, for every fixed* r*,*

$$E^{(r)}_{n,i(n)} = o\big(|D_{n,i(n)}|\big).$$

Proof. Replacing $E^{(r)}_{n,i}$ by $\max_{r'\le r} E^{(r')}_{n,i}$—which only increases the weights, preserves the hypothesis for each fixed r because the maximum of finitely many families, each of total weight $o(N_n)$, has total weight at most their sum, and only strengthens the conclusion—we may and do assume that $E^{(r)}_{n,i}$ is nondecreasing in r.

For $k \in \mathbb{N}$, call an index $i \in I_n$ *k-bad* if $|D_{n,i}| \le k$ or $E^{(k)}_{n,i} > \frac{1}{k}|D_{n,i}|$, and k-good otherwise; let

$$q_k(n) = \frac{1}{N_n} \sum_{i\ k\text{-bad}} |D_{n,i}|.$$

For each fixed k,

$$q_k(n) \le \frac{1}{N_n}\Big(\sum_{i:\,|D_{n,i}|\le k} |D_{n,i}| \;+\; k \sum_{i\in I_n} E^{(k)}_{n,i}\Big) \xrightarrow[n\to\infty]{} 0,$$

the second summand bounding the mass of the second kind of badness by Markov's inequality, and both summands being $o(N_n)$ by hypothesis. Moreover the set of k-bad indices is nondecreasing in k: $|D_{n,i}| \le k$ implies $|D_{n,i}| \le k + 1$, and $E^{(k)}_{n,i} > \frac{1}{k}|D_{n,i}|$ implies $E^{(k+1)}_{n,i} \ge E^{(k)}_{n,i} > \frac{1}{k+1}|D_{n,i}|$, using monotonicity in r. Hence $q_k(n)$ is nondecreasing in k for each fixed n.

Now make one simultaneous choice. Since $q_k(n) \to 0$ as $n \to \infty$ for each fixed k, there are integers $n_1 < n_2 < \cdots$ with $q_k(n) \le \frac{1}{k}$ for all $n \ge n_k$. For

$n \geq n_1$ put $k_n = \max\{k \mid n_k \leq n\}$, a finite maximum since the n_k strictly increase; then $k_n \to \infty$ and, by the definition of n_{k_n}, $q_{k_n}(n) \leq \frac{1}{k_n} \to 0$.

For n large, the k_n-bad indices carry mass $q_{k_n}(n)N_n \leq N_n/k_n = o(N_n)$, while all indices together carry mass at least $(1 - o(1))N_n$; hence some k_n-good index $i(n)$ exists. Then $|D_{n,i(n)}| > k_n \to \infty$, and for every fixed r: as soon as $k_n \geq r$, monotonicity in r gives

$$E^{(r)}_{n,i(n)} \leq E^{(k_n)}_{n,i(n)} \leq \tfrac{1}{k_n}|D_{n,i(n)}| = o\big(|D_{n,i(n)}|\big). \qquad \square$$

3.5. **Localization by completion.**

Localization
`exists_completion`
`involutiveCompletion`
LocalizedApproximation
`toSoficApproximation`

Lemma 3.5 (Localization by completion)**.** *Let H be a countable group and $\sigma_n : H \to \mathrm{Sym}(Y_n)$ maps. Suppose $D_n \subseteq Y_n$ are subsets with $|D_n| \to \infty$ such that, for every fixed $g, h \in H$:*

(i) $|\{y \in D_n \mid \sigma_n(g)y \notin D_n\}| = o(|D_n|)$;
(ii) $|\{y \in D_n \mid \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y\}| = o(|D_n|)$;
(iii) $|\{y \in D_n \mid \sigma_n(g)y = y\}| = o(|D_n|)$ *whenever* $g \neq 1$.

For $g \in H$ define $\pi_n(g) \in \mathrm{Sym}(D_n)$ by letting $\pi_n(g)$ agree with $\sigma_n(g)$ on $\{y \in D_n \mid \sigma_n(g)y \in D_n,\ \sigma_n(g)y \neq \sigma_n(g)y'\ \forall y' \neq y \in D_n\}$ and extending the resulting partial injection of D_n into itself to a permutation of D_n arbitrarily. Then $\pi_n : H \to \mathrm{Sym}(D_n)$ is a sofic approximation of H. Moreover, if $g, g^{-1} \in H$ satisfy $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$, the completions may be chosen with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$, and then

$$\big|\{y \in D_n \mid \pi_n(g)y \neq \sigma_n(g)y\}\big| = o(|D_n|).$$

Proof. Since $\sigma_n(g)$ is injective on Y_n, the described set consists precisely of the points of D_n sent into D_n by $\sigma_n(g)$, and its complement in D_n has size $o(|D_n|)$ by (i); the partial map is an injection, and any extension to a permutation of D_n differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n. Then (ii) transfers: for fixed g, h, the points of D_n where $\pi_n(g)\pi_n(h)y \neq \pi_n(gh)y$ are contained in the union of the $o(|D_n|)$-sized exceptional sets for g, h, gh (points where $\pi \neq \sigma$, including the $\sigma_n(h)$-preimage of the exceptional set of g) and the set in (ii); similarly (iii) transfers faithfulness. When $\sigma_n(g^{-1}) = \sigma_n(g)^{-1}$ and $g \neq g^{-1}$, the two partial injections attached to g and to g^{-1} are mutually inverse, so a completion of one can be transported to the other. When $g = g^{-1}$, so that $\sigma_n(g)^2 = \mathrm{id}$, the set $A_{n,g} = \{y \in D_n \mid \sigma_n(g)y \in D_n\}$ is $\sigma_n(g)$-invariant—if y and $\sigma_n(g)y$ both lie in D_n, then so do $\sigma_n(g)y$ and $\sigma_n(g)^2y = y$—so $\sigma_n(g)$ restricts to an involution of $A_{n,g}$; extending by the identity on $D_n \setminus A_{n,g}$ yields an involutive completion $\pi_n(g) = \pi_n(g)^{-1}$ that differs from $\sigma_n(g)$ on $o(|D_n|)$ points of D_n by (i). $\square$

4. The compression–centralizer criterion

This section proves the analytic core of the paper. We first state a local version that isolates the exact graph-theoretic hypotheses; property (T) of the ambient group is a sufficient condition for the second hypothesis, and the local statement records precisely what it is used for.

4.1. **Statement.**

LocalCriterion `LocalCriterionData` *CriterionAssembly* `isLEF_of_soficApproximation`

Theorem 4.1 (Local compression–centralizer theorem)**.** *Let G be a finitely generated group and let $\Gamma \leq G$ be an infinite finitely generated property-(T) subgroup with finite symmetric generating set S. Let $Q \subseteq G$ be a finite set with*

$$G = \langle \Gamma, Q \rangle \qquad \text{and} \qquad q\Gamma q^{-1} \leq \Gamma \quad (q \in Q). \tag{4}$$

Fix $q_0 \in Q$, put $K = q_0 \Gamma q_0^{-1}$, and let $J \leq \Gamma$ be finitely generated with

$$[K, J] = 1, \qquad K \cap J = \{1\}. \tag{5}$$

Put $A = S \cup Q \cup Q^{-1}$. Suppose G admits a sofic approximation $\sigma_n : G \to \mathrm{Sym}(Y_n)$ such that

(i) *the S-generator graphs $X(Y_n, \sigma_n, S)$ admit an expander decomposition, and*
(ii) *the A-generator graphs $X(Y_n, \sigma_n, A)$ admit an expander decomposition.*

Then J is LEF.

CriterionAssembly `isLEF_of_isSofic` `not_isSofic_of_not_isLEF`

Corollary 4.2 (Kazhdan compression–centralizer criterion; Theorem D)**.** *In the setting of (4) and (5), suppose in addition that G and Γ have property (T). If G is sofic, then J is LEF. Equivalently: if J is not LEF, then G is not sofic.*

Proof. Suppose G is sofic and fix a sofic approximation σ_n. By (4) the finite symmetric set $A = S \cup Q \cup Q^{-1}$ generates G, and G is infinite, containing the infinite subgroup Γ. The restriction $\sigma_n|_\Gamma$ is a sofic approximation of the infinite Kazhdan group Γ, so Theorem 2.9 yields hypothesis (i); applying Theorem 2.9 to G itself with the generating set A yields hypothesis (ii). Theorem 4.1 concludes. □

Remark 4.3. Only the componentwise ℓ^1 median inequality of Lemma 3.3 is extracted from hypothesis (ii); that hypothesis may accordingly be weakened to any decomposition of the A-generator graphs, up to $o(|Y_n|)$ edge edits, into components satisfying a uniform ℓ^1 Poincaré inequality around medians. This separates the approximation-theoretic input, which property (T) supplies through Kun's theorem, from the compression mechanism, which is combinatorial.

The remainder of the section proves Theorem 4.1. Write $N_n = |Y_n|$. All $o(N_n)$ estimates are uniform over the fixed finite data (S, Q, q_0) and the decomposition constants; the set Q itself is never symmetrized inside the compression steps, since the compression hypothesis $q\Gamma q^{-1} \leq \Gamma$ is assumed only for $q \in Q$.

4.2. **The two component partitions.** Put $S_K = q_0 S q_0^{-1}$, a finite symmetric generating set of K, and fix a finite symmetric generating set S_J of the finitely generated group J. By Lemma 2.2 we may and do assume the exact identities

$$\sigma_n(1) = \mathrm{id}, \qquad \sigma_n(g^{-1}) = \sigma_n(g)^{-1} \quad (g \in A \cup S_K \cup S_J),$$

after a pointwise $o(N_n)$ modification; each relevant generator graph changes by $o(N_n)$ edges, so hypotheses (i) and (ii) persist.

By hypothesis (i), fix an expander decomposition of the S-generator graphs: graphs $\widetilde{X}_n$ on Y_n of degree at most d_Γ, with

$$\mathrm{ed}\big(X(Y_n, \sigma_n, S), \widetilde{X}_n\big) = o(N_n),$$

such that every component of $\widetilde{X}_n$ has Cheeger constant at least $h_\Gamma > 0$. Let $\mathcal{P}_n$ denote the partition of Y_n into the components of $\widetilde{X}_n$, and write $P_n(y)$ for the block containing y. We refer to the blocks of $\mathcal{P}_n$ as Γ-*components*.

By hypothesis (ii), fix likewise an expander decomposition $\widetilde{Z}_n$ of the A-generator graphs, with all components of Cheeger constant at least $h_G > 0$ and uniformly bounded degree; let $\mathcal{B}_n$ be the partition of Y_n into the components of $\widetilde{Z}_n$, with blocks $B_n(y)$, the G-*components*.

Both partitions are almost invariant in the following sense. For every $s \in S$,

$$\big|\{y \in Y_n \mid P_n(\sigma_n(s)y) \neq P_n(y)\}\big| = o(N_n), \tag{6}$$

because apart from the $o(N_n)$ edge edits, every arc $(y, \sigma_n(s)y)$ is an edge of $\widetilde{X}_n$ or a loop, and edges of $\widetilde{X}_n$ do not cross $\mathcal{P}_n$. For every $a \in A$,

$$\big|\{y \in Y_n \mid B_n(\sigma_n(a)y) \neq B_n(y)\}\big| = o(N_n), \tag{7}$$

for the same reason applied to $\widetilde{Z}_n$.

Both partitions stay fixed from here on: $\mathcal{P}_n$ and $\mathcal{B}_n$ are the exact component partitions of $\widetilde{X}_n$ and $\widetilde{Z}_n$, and every block keeps its connectivity and its Cheeger constant. This is deliberate—the final step of the proof selects a block of $\mathcal{P}_n$ and uses precisely that it is a genuine component of $\widetilde{X}_n$. The two partitions are related only pointwise, in two ways: through the almost-invariance estimates (6) and (7), and through the median normalization of Section 4.4, which compares a point with its compressor image exactly when the two lie in the same G-component; the $o(N_n)$ points where this fails are carried as explicit exceptional sets through every estimate below. In particular, neither partition is assumed to refine the other.

4.3. **One-sided refinement along a compressor.** Fix $q \in Q$. The group identity $qs = (qsq^{-1})q$ and approximate multiplicativity give, for each $s \in S$, the *conjugacy defect* bound

$$\Big|\Big\{y \in Y_n \,\Big|\, \sigma_n(q)\sigma_n(s)y \neq \sigma_n(qsq^{-1})\sigma_n(q)y\Big\}\Big| = o(N_n). \tag{8}$$

Indeed, $\sigma_n(q)\sigma_n(s)$ agrees with $\sigma_n(qs)$ outside $o(N_n)$ points, and $\sigma_n(qs) = \sigma_n\big((qsq^{-1})q\big)$ agrees with $\sigma_n(qsq^{-1})\sigma_n(q)$ outside $o(N_n)$ points, both by

Definition 2.1(a); here qsq^{-1} belongs to the fixed finite set $qSq^{-1} \subseteq \Gamma$, by the compression hypothesis (4).

Transport the graphs $\widetilde{X}_n$ by the permutation $\sigma_n(q)$: the transported graph $\sigma_n(q)\widetilde{X}_n$ has vertex set Y_n and one edge $\{\sigma_n(q)x, \sigma_n(q)y\}$ for each edge $\{x, y\}$ of $\widetilde{X}_n$, multiplicities preserved; its components are $\{\sigma_n(q)C \mid C \in \mathcal{P}_n\}$, with the same component Cheeger constants $\geq h_\Gamma$ and degree bound d_Γ. By (8) and the edit bounds, apart from $o(N_n)$ edge occurrences the edges of the transported graph are of the form $\{\sigma_n(q)y, \sigma_n(\gamma)\sigma_n(q)y\}$ with $\gamma \in qSq^{-1} \subseteq \Gamma$; for each of the finitely many $\gamma \in qSq^{-1} \subseteq \Gamma$, fix a word in S representing it; by Lemma 2.3, $\sigma_n(\gamma)$ agrees with the product of the letter permutations outside $o(N_n)$ points, and by (6) applied to each letter in turn, that product preserves the block $P_n(\cdot)$ outside $o(N_n)$ points. Hence all but $o(N_n)$ of the transported edges lie inside blocks of $\mathcal{P}_n$.

Apply Lemma 3.1 to the transported graph with the partition $\mathcal{P}_n$. This produces a map

$$\Phi_{q,n} : \mathcal{P}_n \to \mathcal{P}_n, \quad \Phi_{q,n}(C) = \text{the block meeting } \sigma_n(q)C \text{ in maximal size},$$

with total leakage

$$\sum_{C \in \mathcal{P}_n} e_{q,n}(C) = o(N_n), \qquad \text{where} \qquad e_{q,n}(C) := \left|\sigma_n(q)C \setminus \Phi_{q,n}(C)\right|. \tag{9}$$

Thus each transported Γ-component is almost contained in a single target Γ-component. The containment is a priori one-sided: a large target could still absorb several small transported components. Ruling this out is the business of the median normalization.

4.4. **Median normalization and conservation.** For $y \in Y_n$, let $s_n(y) = |P_n(y)|$ be the size of its Γ-component. For a G-component $B \in \mathcal{B}_n$, let $m_n(B) \geq 1$ be a *vertex-weighted median* of s_n on B: a value such that

$$|\{y \in B \mid s_n(y) > m_n(B)\}| \leq \tfrac{1}{2}|B| \quad \text{and} \quad |\{y \in B \mid s_n(y) < m_n(B)\}| \leq \tfrac{1}{2}|B|.$$

Define the *median normalization*

$$f_n(y) = \frac{s_n(y)}{s_n(y) + m_n(B_n(y))} \ \in \ (0,1). \tag{10}$$

Since $t \mapsto t/(t+m)$ is strictly increasing, on every G-component B at most half the vertices satisfy $f_n > \frac{1}{2}$ and at most half satisfy $f_n < \frac{1}{2}$; that is, $\frac{1}{2}$ is a median of f_n on every component of $\widetilde{Z}_n$.

Variation along S. For $s \in S$,

$$\sum_{y \in Y_n} |f_n(\sigma_n(s)y) - f_n(y)| = o(N_n), \tag{11}$$

since the summand vanishes unless $P_n(\sigma_n(s)y) \neq P_n(y)$ or $B_n(\sigma_n(s)y) \neq B_n(y)$, a set of size $o(N_n)$ by (6) and (7), and $0 < f_n < 1$.

One-sided variation along $q \in Q$. Call $y \in Y_n$ a *good point for q* if

$$\sigma_n(q)y \in \Phi_{q,n}(P_n(y)) \qquad \text{and} \qquad B_n(\sigma_n(q)y) = B_n(y). \tag{12}$$

Let y be good, $C = P_n(y)$, $D = \Phi_{q,n}(C)$, and $e = e_{q,n}(C)$; then

$$s_n(\sigma_n(q)y) = |D| \geq |\sigma_n(q)C \cap D| = |C| - e = s_n(y) - e.$$

Since both y and $\sigma_n(q)y$ lie in the same G-component B, they are normalized by the same median $m = m_n(B) \geq 1$. For $a \geq (1-\delta)a' \geq 0$ and $m \geq 1$,

$$\frac{a'}{a'+m} - \frac{a}{a+m} \;\leq\; \frac{a'}{a'+m} - \frac{(1-\delta)a'}{(1-\delta)a'+m} \;\leq\; \delta, \tag{13}$$

where the first inequality is monotonicity of $t \mapsto t/(t+m)$ and the second is a direct computation: the difference equals $\delta a'm/((a'+m)((1-\delta)a'+m)) \leq \delta$. Applying (13) with $a' = |C|$, $a = |D| \geq |C| - e$, $\delta = e/|C|$ gives, for every good point y,

$$\big(f_n(y) - f_n(\sigma_n(q)y)\big)_+ \leq \frac{e_{q,n}(P_n(y))}{|P_n(y)|}.$$

Summing over Y_n—good points of a block C contribute at most $e_{q,n}(C)$ in total, points with $\sigma_n(q)y \notin \Phi_{q,n}(P_n(y))$ number $e_{q,n}(C)$ per block and contribute at most 1 each, and points whose arc crosses $\mathcal{B}_n$ contribute at most 1 each—we obtain

(14)
$$\begin{aligned}\sum_{y \in Y_n} \big(f_n(y) - f_n(\sigma_n(q)y)\big)_+ \;&\leq\; 2\sum_{C \in \mathcal{P}_n} e_{q,n}(C) + \big|\{y \mid B_n(\sigma_n(q)y) \neq B_n(y)\}\big| \\ &=\; o(N_n)\end{aligned}$$

by (9) and (7). The compressed image can only grow—but growth is globally impossible:

Conservation. Since $\sigma_n(q)$ is a permutation of Y_n, Lemma 3.2 upgrades (14) to the two-sided estimate

(15)
$$\sum_{y \in Y_n} |f_n(\sigma_n(q)y) - f_n(y)| = 2\sum_{y \in Y_n} \big(f_n(y) - f_n(\sigma_n(q)y)\big)_+ = o(N_n) \qquad (q \in Q),$$

and the change of variables $y = \sigma_n(q)x$, legitimate because $\sigma_n(q^{-1}) = \sigma_n(q)^{-1}$, extends (15) to all $q \in Q \cup Q^{-1}$.

Pinning. Every edge of the A-generator graph $X(Y_n, \sigma_n, A)$ arises from an arc $(y, \sigma_n(a)y)$ with $a \in A = S \cup Q \cup Q^{-1}$, and each edge arises from at most $2|A|$ arcs. Hence (11) and (15) give

$$\sum_{\{x,y\} \in E(X(Y_n,\sigma_n,A))} |f_n(x) - f_n(y)| = o(N_n).$$

Since $0 < f_n < 1$, editing $o(N_n)$ edges changes this sum by at most $o(N_n)$; therefore the same bound holds over $E(\widetilde{Z}_n)$. As $\frac{1}{2}$ is a median of f_n on every component of $\widetilde{Z}_n$ and all component Cheeger constants are at least h_G, Lemma 3.3 yields the concentration estimate

$$\sum_{y \in Y_n} \left| f_n(y) - \tfrac{1}{2} \right| = o(N_n). \tag{16}$$

In words: after median normalization, almost every Γ-component has size asymptotically equal to the median size on its G-component, because $s/(s+m) \to \frac{1}{2}$ exactly when $s/m \to 1$.

4.5. Almost-bijective matching.

Proposition 4.4 (Matching)**.** *Fix $q \in Q$. There are subcollections $\mathcal{G}_{q,n} \subseteq \mathcal{P}_n$ such that:* MedianNormalization `medianNormalize` MatchingPreparation `acceptable_symmDiff_le` Criterion `symmDiff_le_of_pinned` `matching_injective`

(i) $\sum_{C \in \mathcal{P}_n \setminus \mathcal{G}_{q,n}} |C| = o(N_n)$;
(ii) *for every $C \in \mathcal{G}_{q,n}$, with $D = \Phi_{q,n}(C)$,*

$$\big|\sigma_n(q)C \mathbin{\triangle} D\big| = o(1) \cdot |C| \qquad \textit{uniformly over } C \in \mathcal{G}_{q,n}; \tag{17}$$

(iii) *for n large, $\Phi_{q,n}$ is injective on $\mathcal{G}_{q,n}$;*
(iv) $\sum_{C \in \mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1 - o(1))\, N_n$.

Proof. Choose $\eta_n \downarrow 0$ so slowly that the three global errors—in (7) for the element q, in (9), and in (16)—are all $o(\eta_n^2 N_n)$. Let $W_n \subseteq Y_n$ be the union of the four sets

$$\Big\{y \,\Big|\, |f_n(y) - \tfrac{1}{2}| > \eta_n\Big\}, \quad \sigma_n(q)^{-1} \Big\{y \,\Big|\, |f_n(y) - \tfrac{1}{2}| > \eta_n\Big\},$$
$$\{y \,|\, B_n(\sigma_n(q)y) \neq B_n(y)\}, \quad \{y \,|\, \sigma_n(q)y \notin \Phi_{q,n}(P_n(y))\}.$$

Markov's inequality applied to (16) bounds the first set by $\eta_n^{-1} \cdot o(\eta_n^2 N_n) = o(\eta_n N_n)$; the second is a bijective preimage of the first; the last two have sizes $o(\eta_n^2 N_n)$ by the choice of η_n. Hence $|W_n| = o(\eta_n N_n)$.

Discard from $\mathcal{P}_n$ every block C with $e_{q,n}(C) > \eta_n|C|$ or $|C \cap W_n| > \eta_n|C|$, and call the remaining collection $\mathcal{G}_{q,n}$. The discarded mass is at most $\eta_n^{-1} \sum_C e_{q,n}(C) + \eta_n^{-1}|W_n| = o(N_n)$, proving (i).

Let $C \in \mathcal{G}_{q,n}$ and pick $y \in C \setminus W_n$, possible since $\eta_n < 1$. Put $D = \Phi_{q,n}(C)$ and $B = B_n(y) = B_n(\sigma_n(q)y)$, and write $m = m_n(B)$. Since $y \notin W_n$, we have $\sigma_n(q)y \in D$, so $s_n(\sigma_n(q)y) = |D|$, and

$$\Big|\frac{|C|}{|C| + m} - \frac{1}{2}\Big| \le \eta_n, \qquad \Big|\frac{|D|}{|D| + m} - \frac{1}{2}\Big| \le \eta_n.$$

Solving $|\frac{a}{a+m} - \frac{1}{2}| \le \eta$ gives $\frac{1-2\eta}{1+2\eta}\, m \le a \le \frac{1+2\eta}{1-2\eta}\, m$; applying this to $a = |C|$ and to $a = |D|$ and dividing,

$$|D| = (1 + O(\eta_n))\, |C| \qquad \text{uniformly over } C \in \mathcal{G}_{q,n}. \tag{18}$$

With $e = e_{q,n}(C) \le \eta_n|C|$,

$$\begin{aligned} |\sigma_n(q)C \mathbin{\triangle} D| &= |\sigma_n(q)C \setminus D| + |D \setminus \sigma_n(q)C| \\ &\le e + (|D| - |C| + e) \le 2e + \big||D| - |C|\big| = O(\eta_n)|C|, \end{aligned}$$

proving (ii).

For (iii): if $C \neq C'$ in $\mathcal{G}_{q,n}$ had $\Phi_{q,n}(C) = \Phi_{q,n}(C') = D$, the disjoint sets $\sigma_n(q)C \cap D$ and $\sigma_n(q)C' \cap D$ would, by (17) and (18), each occupy a fraction $1 - o(1)$ of D, uniformly—impossible for n large.

For (iv): by (i), (ii), and (18),

$$\sum_{C\in\mathcal{G}_{q,n}} |\Phi_{q,n}(C)| = (1+o(1)) \sum_{C\in\mathcal{G}_{q,n}} |C| = (1+o(1))(1-o(1))N_n = (1-o(1))N_n.$$

□

Thus one-sided refinement, conservation, and pinning together force the compressor to permute the Γ-components almost bijectively, with asymptotically equal sizes. One more observation is needed before selecting a single matched pair.

ComponentDivergence `smallBlockVertices_negligible`

Lemma 4.5. *For every fixed M, $\sum\{|C| : C \in \mathcal{P}_n,\ |C| \le M\} = o(N_n)$.*

Proof. Let $g_1, \dots, g_{M+1} \in \Gamma$ be distinct elements; they exist since Γ is infinite. For all but $o(N_n)$ points y the points $\sigma_n(g_1)y, \dots, \sigma_n(g_{M+1})y$ are pairwise distinct: for $i \neq j$, $\sigma_n(g_i)$ agrees with $\sigma_n(g_i g_j^{-1})\sigma_n(g_j)$ outside $o(N_n)$ points by Definition 2.1(a), and $\sigma_n(g_i g_j^{-1})$ moves $1 - o(1)$ of all points by Definition 2.1(b). Moreover, fixing a word in S representing each g_i, Lemma 2.3 together with (6) applied along its letters shows that for all but $o(N_n)$ points y all these points lie in $P_n(y)$. A point y satisfying both conditions has $|P_n(y)| \ge M + 1$; hence the mass of blocks of size at most M is $o(N_n)$. □

4.6. **Selection, localization, and the proof of Theorem 4.1.**

Proof of Theorem 4.1. Apply Proposition 4.4 with $q = q_0$. For n large this gives a collection of *matched pairs*

$$\big\{(C, D) : C \in \mathcal{G}_{q_0,n},\ D = \Phi_{q_0,n}(C)\big\}, \tag{19}$$

with pairwise distinct sources, pairwise distinct targets, target mass $(1 - o(1))N_n$, and $|\sigma_n(q_0)C \,\triangle\, D| = o(1)|C|$ uniformly.

Fix an exhaustion $L_1 \subseteq L_2 \subseteq \cdots$ of the countable group $KJ \le \Gamma$ by finite subsets with $\bigcup_r L_r = KJ$, each L_r containing 1, the finite sets S_K and S_J, and closed under inverses, and such that $L_r \cdot L_r \subseteq L_{r+1}$. To each matched pair (C, D) and each r attach the error weight $E^{(r)}(C, D)$, the sum of:

(a) $|\sigma_n(q_0)C \,\triangle\, D|$;
(b) the number of edges of $\widetilde{X}_n$ with an endpoint in C or D that are edited relative to the S-generator graph, plus the number of edges of the transported graph $\sigma_n(q_0)\widetilde{X}_n$ with an endpoint in $D \cup \sigma_n(q_0)C$ that are edited relative to the transported S-generator graph;
(c) $\left|\left\{y \in C \,\middle|\, \sigma_n(q_0)\sigma_n(s)y \neq \sigma_n(q_0 s q_0^{-1})\sigma_n(q_0)y \text{ for some } s \in S\right\}\right|$;
(d) $|\{y \in D \,|\, \sigma_n(g)\sigma_n(h)y \neq \sigma_n(gh)y \text{ for some } g, h \in L_r\}|$;
(e) $|\{y \in D \,|\, \sigma_n(g)y = y \text{ for some } g \in L_r \setminus \{1\}\}|$;
(f) $|\{y \in D \,|\, P_n(\sigma_n(g)y) \neq P_n(y) \text{ for some } g \in L_r\}|$.

For every fixed r, the sum of $E^{(r)}$ over the matched pairs is $o(N_n)$: (a) by Proposition 4.4(ii) and (i); (b): the sources are pairwise disjoint, the targets are pairwise disjoint, and a fixed block can occur at most once as a source and at most once as a target; hence each endpoint of an edge lies in at most one source and at most one target, and, an edge having two endpoints, each edited edge is counted for at most four matched pairs—likewise in the transported graph, where the sets $\sigma_n(q_0)C$ are pairwise disjoint images of the disjoint sources—while the global edit counts are $o(N_n)$; (c) by (8) over the finite set S, the sources being disjoint; (d) and (e) by soficity of σ_n over the finite set $L_r \cup L_r^2$, the targets being disjoint; (f) by Lemma 2.3 together with (6) applied along the letters of a fixed word in S representing each $g \in L_r \subseteq \Gamma$, the targets being disjoint.

The targets D are pairwise disjoint blocks of $\mathcal{P}_n$ with total mass $(1 - o(1))N_n$, and by Lemma 4.5 the total mass of targets of size at most M is $o(N_n)$ for every fixed M. Lemma 3.4 therefore produces matched pairs (C_n, D_n) with

$$|D_n| \to \infty, \qquad E^{(r)}(C_n, D_n) = o(|D_n|) \quad \text{for every fixed } r. \tag{20}$$

Put $U_n = \sigma_n(q_0)C_n$, so $|U_n \triangle D_n| = o(|D_n|)$ and $|U_n| = |C_n| = (1+o(1))|D_n|$. Define

$$D_n' = U_n, \tag{21}$$

a set of size $|C_n| \to \infty$ that agrees with D_n up to $o(|D_n|)$ points. Every error weight in (20) transfers from the pair (C_n, D_n) to D_n' at a cost of $o(|D_n'|)$, by separate accounting: the point-counted errors (d), (e), (f) are counted over D_n, and replacing D_n by D_n' changes each count by at most $|D_n \triangle D_n'| = o(|D_n'|)$; the error (c) is counted over C_n, which is unchanged; the edge-counted errors (b) concern graphs of degree at most $\max(d_\Gamma, 2|S|)$, so replacing D_n by D_n' in the incidence condition changes those counts by at most that degree bound times $|D_n \triangle D_n'| = o(|D_n'|)$; and the quantity (a) does not change.

Localization. Consider the countable group $H = KJ \le \Gamma$. By (5), K and J commute and intersect trivially, so

$$H = KJ \;\cong\; K \times J.$$

For every fixed $g \in H$, the transferred error (f) shows that $\sigma_n(g)$ maps all but $o(|D_n'|)$ points of D_n' into D_n. Injectivity upgrades D_n to D_n': since $\sigma_n(g)$ is a permutation of Y_n, at most $|D_n \setminus D_n'| = o(|D_n'|)$ points of D_n' can land in $D_n \setminus D_n'$, so $\sigma_n(g)$ maps all but $o(|D_n'|)$ points of D_n' into D_n' itself. The transferred errors (d) and (e) give multiplicativity and faithfulness on D_n' for every fixed pair $g, h \in H$, by choosing r with $g, h \in L_r$. Lemma 3.5, with completions on the symmetric set $S_K \cup S_J$ chosen compatibly with the exact inverse identities of Section 4.2, yields a sofic approximation

$$\pi_n : K \times J \longrightarrow \mathrm{Sym}(D_n')$$

with $|\{y \in D_n' \mid \pi_n(g)y \neq \sigma_n(g)y\}| = o(|D_n'|)$ for every fixed g, and $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K \cup S_J$.

The K-edges expand. Restrict $\widetilde{X}_n$ to C_n and transport by $\sigma_n(q_0)$: the resulting graph Ξ_n on $D_n' = \sigma_n(q_0)C_n$ is connected with Cheeger constant at least h_Γ and degree at most d_Γ, because C_n is a component of $\widetilde{X}_n$. We claim that Ξ_n and the S_K-generator graph $X(D'_n, \pi_n, S_K)$ differ by $o(|D'_n|)$ edges. Indeed, an edge of either graph that fails to be an edge of the other must involve one of the following, each affecting $o(|D'_n|)$ edge occurrences by (20):

(1) an edge of $\widetilde{X}_n[C_n]$ edited relative to the S-generator graph on C_n, or conversely (error (b));
(2) an arc $(y, \sigma_n(s)y)$, $y \in C_n$, whose $\sigma_n(q_0)$-transport disagrees with the $\sigma_n(q_0 s q_0^{-1})$-arc at $\sigma_n(q_0)y$ (error (c));
(3) a point of D'_n where $\pi_n(g) \neq \sigma_n(g)$ for some $g \in S_K$ (localization defect);
(4) an arc of some $g \in S_K$ leaving D'_n (transferred error (f) and the symmetric difference (a)).

Hence the S_K-generator graphs of $\pi_n|_K$ form an essential expander sequence in the sense of Definition 2.6.

Conclusion. The group $K = q_0 \Gamma q_0^{-1} \cong \Gamma$ is an infinite finitely generated property-(T) group with symmetric generating set S_K, the group J is finitely generated, and π_n is a sofic approximation of $K \times J$ whose S_K-generator graphs form an essential expander sequence. The regular-labeling requirements recorded after Theorem 2.10 hold for π_n as well: each $\pi_n(g)$ is a genuine permutation of D'_n, with $\pi_n(g^{-1}) = \pi_n(g)^{-1}$ for $g \in S_K \cup S_J$; and for $g, h \in S_K \cup S_J$ with $h \neq g^{\pm 1}$, the collision set $\{y \in D'_n \mid \pi_n(g)y = \pi_n(h)y\}$ has size $o(|D'_n|)$, by the transferred errors (d) and (e) applied to the pair (h^{-1}, g) and the element $h^{-1}g \in L_2 \setminus \{1\}$, together with the $o(|D'_n|)$ agreement between π_n and σ_n on $S_K \cup S_J \cup \{h^{-1}g\}$. Theorem 2.10 applies and shows that J is LEF. □

4.7. **Two remarks on the mechanism.**

Remark 4.6 (The product obstruction)**.** The compression hypothesis cannot be deleted from Theorem D. For a direct product $\Lambda \times \Delta$ of an infinite Kazhdan group with a residually finite Kazhdan group, product sofic approximations exist whenever Λ is sofic, and in them the Δ-factor genuinely permutes the Λ-components: the Λ-component partition has all blocks of equal size, and no single component is almost invariant under the second factor. The role of the compressors $q \in Q$, through (4), is to generate the whole ambient group while conjugating Γ *into* itself, so that the ambient expansion of hypothesis (ii) couples the component structure of Γ to itself across all of G and forbids exactly this shuffling.

Remark 4.7 (The hypothesis $J \leq \Gamma$ is essential)**.** The subgroup J is required to lie in Γ, and this is used twice: through (6) in error (f), which keeps the J-arcs inside the matched component, and through the localization step, which restricts σ_n on elements of $KJ \leq \Gamma$. The hypothesis cannot be dropped in favor of $J \leq G$. Indeed, groups satisfying all the remaining hypotheses with a non-LEF subgroup $J \leq G$ commuting with K exist in abundance among sofic groups—already $J = \mathrm{BS}(2,3) = \langle a, t \mid ta^2t^{-1} = a^3 \rangle$

is a finitely presented group that is known to be sofic and is non-Hopfian [4], therefore not residually finite, therefore not LEF by Lemma 2.8. The criterion is calibrated exactly so that the commuting subgroup rides inside the compressed expander, and this is what the Leavitt corner construction of Sections 6–7 arranges algebraically.

5. Leavitt algebras, elementary groups, and Thompson's group V

Throughout, k is a field and $L = L_k(1,2)$ is the binary Leavitt algebra (1), with generators s_0, s_1, t_0, t_1 and relations $t_i s_j = \delta_{ij}$, $s_0t_0 + s_1t_1 = 1$. Everything in this section is a consequence of the two relations alone, and it costs nothing to say so once and for all.

Definition 5.1. A *binary Leavitt family* in a unital ring A is a quadruple (s_0, s_1, t_0, t_1) of elements of A satisfying Leavitt LeavittFamily

$$t_is_j = \delta_{ij} \quad (i,j \in \{0,1\}) \qquad \text{and} \qquad s_0t_0 + s_1t_1 = 1.$$

The generators of L form a binary Leavitt family in L, and a unital homomorphic image of L carries one; but a family is strictly less than a copy of L—no injectivity is assumed—and the statements below that mention only a family are correspondingly stronger than their specializations to L. Fix a unital ring A carrying a binary Leavitt family, and write

$$p_0 = s_0t_0, \qquad p_1 = s_1t_1,$$

a pair of orthogonal idempotents summing to 1: $p_ip_j = \delta_{ij}p_i$, since $p_0p_1 = s_0(t_0s_1)t_1 = 0$ and $p_i^2 = s_i(t_is_i)t_i = p_i$. For a binary word $w = w_1 \cdots w_\ell \in \{0,1\}^*$, put $s_w = s_{w_1} \cdots s_{w_\ell}$ and $s_w^* = t_{w_\ell} \cdots t_{w_1}$, so that $s_w^* s_{w'} = \delta_{w,w'}$ whenever $|w| = |w'|$, and more generally $s_w^* s_{w'}$ vanishes unless one of w, w' is a prefix of the other.

Two remarks on size. The algebra L itself is infinite-dimensional over k: the elements $s_{0^i}s_1$, $i \geq 0$, are killed on the left by mutually orthogonal projections and are linearly independent. And a family forces infinitude with no reference to L:

Lemma 5.2. *A nontrivial unital ring A carrying a binary Leavitt family is infinite.* Leavitt infinite s0_pow_injective

Proof. The idempotents $e_i = s_0^it_0^i$, $i \geq 0$, are pairwise distinct. Indeed, suppose $e_i = e_j$ with $i < j$. Multiplying on the left by t_0^i and on the right by s_0^i, and using $t_0^is_0^i = 1$, gives $1 = s_0^dt_0^d$ with $d = j - i \geq 1$. Then $t_1 = t_1 \cdot 1 = (t_1s_0)\, s_0^{d-1}t_0^d = 0$, whence $1 = t_1s_1 = 0$, contradicting nontriviality. $\square$

5.1. Leaf sets and matrix self-similarity.

5.1. Leaf sets and matrix self-similarity. A finite set of binary words $\mathcal{C} = (\alpha_1, \ldots, \alpha_r)$ (we use the same symbols for the words and for the corresponding elements $s_{\alpha_i} \in A$) is a *complete leaf set* if the words are the leaves of a finite full rooted binary tree; equivalently, if they are pairwise prefix-incomparable and the cylinders they define partition the boundary $\{0,1\}^{\mathbb{N}}$.

LeavittWords
`wordT_mul_wordS_self`
`wordT_mul_wordS_of_incomparable`
`cylinder_split`

Lemma 5.3 (Leaf identities)**.** *Let $\mathcal{C} = (\alpha_1, \ldots, \alpha_r)$ be a complete leaf set. Then in every unital ring A carrying a binary Leavitt family:*

$$\alpha_i^* \alpha_j = \delta_{ij} \qquad \text{and} \qquad \sum_{i=1}^{r} \alpha_i \alpha_i^* = 1.$$

Proof. Prefix-incomparability and the relations $t_i s_j = \delta_{ij}$ give the first identity. The second holds by induction on the tree: for the trivial tree it reads $1 = 1$; replacing a leaf α by its two children $\alpha 0, \alpha 1$ changes the sum by

$$s_{\alpha 0} s_{\alpha 0}^* + s_{\alpha 1} s_{\alpha 1}^* - s_\alpha s_\alpha^* = s_\alpha (s_0 t_0 + s_1 t_1 - 1) s_\alpha^* = 0. \qquad \square$$

MatrixSelfSimilarity
`ringEquivMatrix`
LeavittSelfSimilarity
`binaryMatrixRingEquiv`
Every positive rank, with an explicit inverse.

Proposition 5.4 (Matrix self-similarity)**.** *Let A be a unital ring carrying a binary Leavitt family and let $\mathcal{C} = (\alpha_1, \ldots, \alpha_r)$ be an ordered complete leaf set. Then*

$$\Theta_{\mathcal{C}} : M_r(A) \longrightarrow A, \qquad (a_{ij}) \longmapsto \sum_{i,j=1}^{r} \alpha_i \, a_{ij} \, \alpha_j^*,$$

is an isomorphism of unital rings—of unital k-algebras when A is a k-algebra—with inverse $x \mapsto (\alpha_i^* x \alpha_j)_{ij}$. It restricts to group isomorphisms $\mathrm{GL}_r(A) \cong A^\times$ for all $r \geq 1$. In particular $M_r(L) \cong L$ for every $r \geq 1$.

Proof. The map $\Theta_{\mathcal{C}}$ is additive—and k-linear when A is a k-algebra—sends the identity matrix to $\sum_i \alpha_i \alpha_i^* = 1$ by Lemma 5.3, and is multiplicative:

$$\Theta_{\mathcal{C}}(a)\, \Theta_{\mathcal{C}}(b) = \sum_{i,j,l,m} \alpha_i a_{ij} \, \alpha_j^* \alpha_l \, b_{lm} \alpha_m^* = \sum_{i,m} \alpha_i \Big(\sum_j a_{ij} b_{jm} \Big) \alpha_m^* = \Theta_{\mathcal{C}}(ab),$$

using $\alpha_j^* \alpha_l = \delta_{jl}$. The displayed inverse is two-sided by the same identities: $\alpha_i^* \Theta_{\mathcal{C}}(a) \alpha_j = a_{ij}$, and $\sum_{i,j} \alpha_i (\alpha_i^* x \alpha_j) \alpha_j^* = 1 \cdot x \cdot 1 = x$. An algebra isomorphism restricts to an isomorphism of unit groups, and $\mathrm{GL}_r(L) = M_r(L)^\times$. $\square$

For every $r \geq 2$ there is a canonical choice, the *left comb*

$$\mathcal{C}_r = \big(s_1,\ s_0 s_1,\ s_0^2 s_1,\ \ldots,\ s_0^{r-2} s_1,\ s_0^{r-1}\big), \tag{22}$$

the leaf set of the tree in which every left child is subdivided; the ordering lists the leaves by depth with the leftmost branch last.

5.2. **Elementary groups over rings.** Let A be a unital ring. Recall $x_{ij}(a) = I_r + a E_{ij}$ for $i \neq j$, the relations $x_{ij}(a) x_{ij}(b) = x_{ij}(a+b)$, and the Steinberg commutator relation

$$[x_{il}(a), x_{lj}(b)] = x_{ij}(ab) \qquad (i, j, l \text{ distinct}),$$

where $[g,h] = g h g^{-1} h^{-1}$.

Lemma 5.5 (Whitehead commutator lemma)**.** *Let A be a unital ring and $a, b \in A^\times$. Put*

$$w(a) = x_{12}(a)\, x_{21}(-a^{-1})\, x_{12}(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}.$$

Then

$$\operatorname{diag}(a, a^{-1}) = w(a)\, w(-1) \in \mathrm{EL}_2(A),$$

and

$$\operatorname{diag}(aba^{-1}b^{-1},\, 1) = \operatorname{diag}(a, a^{-1})\ \operatorname{diag}(b, b^{-1})\ \operatorname{diag}\big((ba)^{-1}, ba\big) \in \mathrm{EL}_2(A).$$

Consequently, for every $m \geq 2$, every product of commutators of units $c \in [A^\times, A^\times]$, and every diagonal position, the matrix with c at that position and 1 elsewhere on the diagonal lies in $\mathrm{EL}_m(A)$.

Whitehead
`whitehead_commutator`
`whitehead_diagonal`
MatrixDiagonalization
`exists_elementary_whitehead`

Proof. First, $x_{12}(a)x_{21}(-a^{-1}) = \begin{pmatrix} 1-aa^{-1} & a \\ -a^{-1} & 1 \end{pmatrix} = \begin{pmatrix} 0 & a \\ -a^{-1} & 1 \end{pmatrix}$, and multiplying by $x_{12}(a)$ on the right adds the first column, multiplied on the right by a, to the second, giving $w(a) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix}$. Then

$$w(a)w(-1) = \begin{pmatrix} 0 & a \\ -a^{-1} & 0 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix}.$$

The displayed product of three diagonal matrices has first entry $ab(ba)^{-1} = aba^{-1}b^{-1}$ and second entry $a^{-1}b^{-1}(ba) = 1$. For the last statement, embed $\mathrm{EL}_2(A) \hookrightarrow \mathrm{EL}_m(A)$ in any chosen pair of coordinates and multiply the resulting matrices for the commutator factors of c. $\square$

Lemma 5.6. *Let A be a finitely generated unital ring and $r \geq 3$. Then $\mathrm{EL}_r(A)$ is finitely generated: if A is generated as a unital ring by the finite set T, then $\mathrm{EL}_r(A)$ is generated by the finitely many matrices $x_{ij}(1)$ and $x_{ij}(t)$, $t \in T$, $i \neq j$.*

ElementaryGroup
`elementaryGroup_finitelyGenerated`

Proof. Let E be the subgroup generated by the listed matrices, and let $A_0 \subseteq A$ be the set of a with $x_{ij}(a) \in E$ for all $i \neq j$. By the addition relation, A_0 is an additive subgroup containing 1 and T; by the Steinberg relation with a third index $l \notin \{i, j\}$ (here $r \geq 3$ is used), $a, b \in A_0$ implies $ab \in A_0$, since $x_{ij}(ab) = [x_{il}(a), x_{lj}(b)]$ and $x_{il}(a), x_{lj}(b) \in E$. Hence A_0 is a unital subring containing T, so $A_0 = A$ and $E = \mathrm{EL}_r(A)$. $\square$

Theorem 5.7 (Ershov–Jaikin-Zapirain [9])**.** *Let A be a finitely generated associative unital ring and $r \geq 3$. Then $\mathrm{EL}_r(A)$ has Kazhdan's property (T).*

formalized in part
FreeElementaryPropertyT
`controlSet_isKazhdanPair`
FiniteFieldElementaryPropertyT
`finiteFieldElementaryThree_hasKazhdanPropertyT`
Proved internally at rank three for finite-type algebras over every finite field, with an explicit Kazhdan pair; the arbitrary finitely generated ring statement is manuscript-only.

The theorem in [9] is stated for the universal finitely generated rings $\mathbb{Z}\langle x_1, \ldots, x_d \rangle$; the general case follows since EL_r of a quotient ring is a quotient group of EL_r of the ring, elementary generators mapping to elementary generators, and property (T) passes to quotients.

Over a ring carrying a binary Leavitt family, the rank of an elementary group is immaterial: matrix self-similarity collapses all ranks at the level of the *elementary* groups themselves, with no K-theory and no GE input. This is the engine that will spread every property of one rank to all ranks.

LeavittRankEquivalence
`rankSuccEquiv`
`rankSuccToProduct`
`rankSucc_propertyT_of_rankSucc`

Proposition 5.8 (Elementary rank equivalence). *Let A be a unital ring carrying a binary Leavitt family and let $n, m \geq 2$. Then there is an explicit group isomorphism*

$$\mathrm{EL}_n(A) \;\cong\; \mathrm{EL}_m(A),$$

the composite

$$\mathrm{EL}_n(A) \;\cong\; \mathrm{EL}_n(M_m(A)) \;=\; \mathrm{EL}_{nm}(A) \;=\; \mathrm{EL}_m(M_n(A)) \;\cong\; \mathrm{EL}_m(A),$$

where the outer isomorphisms apply $\Theta_{\mathcal{C}}^{-1}$ and $\Theta_{\mathcal{C}'}$ of Proposition 5.4 to each matrix coefficient, and the middle equalities are block flattening. In particular finite generation, infinitude, property (T), and (non)soficity hold at one rank ≥ 2 if and only if they hold at every rank ≥ 2.

Proof. Coefficient transport. A unital ring isomorphism $\varphi : B \to B'$ carries $x_{ij}(b)$ to $x_{ij}(\varphi b)$, hence induces an isomorphism $\mathrm{EL}_p(B) \cong \mathrm{EL}_p(B')$ for every p. Apply this with $\varphi = \Theta_{\mathcal{C}_m}^{-1} : A \to M_m(A)$, which exists by Proposition 5.4.

Block flattening. Identify $M_p(M_q(A)) = M_{pq}(A)$ by the index pairing (i, c), $1 \leq i \leq p$, $1 \leq c \leq q$. We claim $\mathrm{EL}_p(M_q(A)) = \mathrm{EL}_{pq}(A)$ as subgroups of $\mathrm{GL}_{pq}(A)$, for every $p \geq 2$ and $q \geq 1$. One inclusion is direct: a block transvection $I + aE_{ij}$, $i \neq j$, $a \in M_q(A)$, equals the product $\prod_{c,d} x_{(i,c),(j,d)}(a_{cd})$ of q^2 commuting entry transvections. For the other inclusion, an entry transvection $x_{(i,c),(j,d)}(r)$ with $i \neq j$ is a block transvection with the single-entry coefficient rE_{cd}; and one with $i = j$ (so $c \neq d$) is produced by the Steinberg relation through any third block index $l \neq i$, which exists because $p \geq 2$:

$$x_{(i,c),(i,d)}(r) = \big[\, x_{(i,c),(l,c)}(r),\; x_{(l,c),(i,d)}(1) \,\big],$$

both factors being cross-block transvections, hence in $\mathrm{EL}_p(M_q(A))$ by the previous sentence. Applying block flattening with $(p, q) = (n, m)$ and—after reindexing along the bijection $(i, c) \mapsto (c, i)$, which conjugates by a permutation matrix and matches entry transvections to entry transvections—with $(p, q) = (m, n)$, gives the middle equalities. The final claim holds because all four properties are isomorphism-invariant. ☐

5.3. **Characteristic-two block factorizations.**

The following two identities, valid in characteristic two, are the engine of the explicit spine.

Leavitt
`characteristicTwo_involution`
`characteristicTwo_compressor`
`compressor_factorization`
The factorization is proved in every characteristic.

Lemma 5.9. *Let A be a unital ring of characteristic 2 carrying a binary Leavitt family. Then, in $M_2(A)$:*

(a) $x_{12}(s_1)\, x_{21}(t_1)\, x_{12}(s_1) = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}$;

(b) $x_{21}(1 + t_0)\, x_{12}(1)\, x_{21}(1 + s_0)\, x_{12}(t_0) = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix}$.

Proof. (a) Since $1 + p_1 = p_0$ in characteristic 2,

$$\begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} 1 + s_1 t_1 & s_1 \\ t_1 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix},$$

and then

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 1 \end{pmatrix}\begin{pmatrix} 1 & s_1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} p_0 & p_0 s_1 + s_1 \\ t_1 & t_1 s_1 + 1 \end{pmatrix} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix},$$

using $p_0 s_1 = s_0(t_0 s_1) = 0$ and $t_1 s_1 + 1 = 1 + 1 = 0$.

(b) Compute the product of the first two factors:

$$\begin{pmatrix} 1 & 0 \\ 1+t_0 & 1 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1+t_0 & t_0 \end{pmatrix}$$

(using $1 + t_0 + 1 = t_0$ in characteristic 2). Multiplying by the third factor,

$$\begin{pmatrix} 1 & 1 \\ 1+t_0 & t_0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 1+s_0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 1+t_0+t_0+t_0 s_0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix},$$

where the $(1,1)$-entry is $1 + 1 + s_0 = s_0$ and the $(2,1)$-entry is $(1 + t_0) + t_0(1 + s_0) = 1 + t_0 + t_0 + t_0 s_0 = 1 + 1 = 0$, using $t_0 s_0 = 1$. Finally,

$$\begin{pmatrix} s_0 & 1 \\ 0 & t_0 \end{pmatrix}\begin{pmatrix} 1 & t_0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} s_0 & s_0 t_0 + 1 \\ 0 & t_0 \end{pmatrix} = \begin{pmatrix} s_0 & s_1 t_1 \\ 0 & t_0 \end{pmatrix},$$

since $s_0 t_0 + 1 = p_0 + 1 = p_1 = s_1 t_1$ in characteristic 2. □

5.4. **Thompson's group V in a Leavitt corner.** Thompson's group V is the group of homeomorphisms of the boundary $\{0,1\}^{\mathbb{N}}$ given by *tree tables*: a pair of ordered complete leaf sets $(\beta_1, \dots, \beta_\ell)$, $(\alpha_1, \dots, \alpha_\ell)$ with the same number ℓ of leaves, acting by the prefix substitutions $\beta_i x \mapsto \alpha_i x$; each such map carries the cylinder of β_i bijectively onto the cylinder of α_i and is a homeomorphism of the Cantor set. Two tables define the same element exactly when they have a common refinement, refinement being simultaneous subdivision of a source leaf and its target leaf. The group V is infinite, simple, and finitely presented [11, 6], and 2-generated [5].

Proposition 5.10 (Tree-table embedding)**.** *Let A be a nontrivial unital ring carrying a binary Leavitt family. For $g \in V$ given by a table $\big((\beta_i), (\alpha_i)\big)$, put*

ThompsonVEmbedding
`vEmbedding`
`vEmbedding_injective`
`isComplete_of_covers`
ThompsonV
`tableEquiv`

$$U_g = \sum_{i=1}^{\ell} \alpha_i\, \beta_i^* \;\in\; A. \tag{23}$$

Then U_g is well defined (independent of the table), $U_g U_h = U_{gh}$, $U_1 = 1$, each U_g is a unit with $U_g^{-1} = U_{g^{-1}}$, and $g \mapsto U_g$ is an injective homomorphism $V \hookrightarrow A^\times$.

Proof. Refining a table replaces a summand $\alpha\beta^*$ by $\alpha 0\,(\beta 0)^* + \alpha 1\,(\beta 1)^* = \alpha(s_0 t_0 + s_1 t_1)\beta^* = \alpha\beta^*$; since any two tables of the same element admit a common refinement, U_g is well defined. For multiplicativity, choose tables of g and h such that the source leaf set of g equals the target leaf set of h (refine both accordingly); then

$$U_g U_h = \sum_{i,j} \alpha_i \beta_i^*\, \gamma_j \delta_j^* = \sum_i \alpha_i \delta_i^* = U_{gh},$$

using $\beta_i^*\gamma_j = \delta_{ij}$ for the common leaf set, and the composed table represents gh. Also $U_1 = \sum_i \alpha_i\alpha_i^* = 1$ by Lemma 5.3, so each U_g is a unit with inverse $U_{g^{-1}}$. For injectivity, suppose $U_g = 1$ with a table $\big((\beta_i), (\alpha_i)\big)$. Multiplying $\sum_i \alpha_i\beta_i^* = 1$ on the right by β_j gives $s_{\alpha_j} = s_{\beta_j}$ in A for each j—the products $\beta_i^*\beta_j = \delta_{ij}$ collapse the sum. Equality of these elements forces equality of the binary words. If neither word is a prefix of the other, multiplying $s_{\alpha_j} = s_{\beta_j}$ on the left by $s_{\alpha_j}^*$ gives $1 = s_{\alpha_j}^* s_{\beta_j} = 0$, contradicting the nontriviality of A. If, say, $\beta_j = \alpha_j\gamma$ with γ nonempty, the same multiplication gives $1 = s_\gamma$; choosing $i \in \{0, 1\}$ different from the first letter of γ then gives $t_i = t_i\, s_\gamma = 0$, whence $1 = t_i s_i = 0$, again a contradiction. The case $\alpha_j = \beta_j\gamma$ is symmetric. Hence $\alpha_j = \beta_j$ as words for every j, and g is the identity substitution. This is the standard realization of the Higman–Thompson groups inside Leavitt algebras and their Cuntz-algebra completions; compare [16, 17]. □

LeavittCorner
`cornerHom`
`cornerHom_s0`
`t0_cornerHom`
`t1_cornerHom_s1`

Lemma 5.11 (Corner embedding)**.** *Let A be a nontrivial unital ring carrying a binary Leavitt family. For $g \in V$ put*

$$\rho(g) = p_0 + s_1 U_g t_1 \ \in\ A.$$

Then $\rho : V \to A^\times$ is an injective group homomorphism, and for all $g \in V$:

$$t_1\rho(g)s_1 = U_g, \qquad \rho(g)\, s_0 = s_0, \qquad t_0\, \rho(g) = t_0.$$

Proof. All computations take place in the subring generated by the Leavitt family, using $t_1s_0 = 0$, $t_0s_1 = 0$, $t_1s_1 = t_0s_0 = 1$, $p_0s_0 = s_0$, $t_0p_0 = t_0$, $p_0s_1 = 0$, $t_1p_0 = 0$. Multiplicativity:

$$\begin{aligned}\rho(g)\rho(h) &= (p_0 + s_1U_gt_1)(p_0 + s_1U_ht_1)\\ &= p_0^2 + p_0s_1U_ht_1 + s_1U_gt_1p_0 + s_1U_gt_1s_1U_ht_1\\ &= p_0 + s_1U_gU_ht_1 = \rho(gh),\end{aligned}$$

by Proposition 5.10. Also $\rho(1) = p_0 + p_1 = 1$, so $\rho(g) \in A^\times$ with inverse $\rho(g^{-1})$. The three displayed identities are immediate: $t_1\rho(g)s_1 = t_1p_0s_1 + t_1s_1U_gt_1s_1 = U_g$; $\rho(g)s_0 = p_0s_0+s_1U_gt_1s_0 = s_0$; $t_0\rho(g) = t_0p_0+t_0s_1U_gt_1 = t_0$. Injectivity: $\rho(g) = 1$ forces $U_g = t_1\rho(g)s_1 = t_1s_1 = 1$, hence $g = 1$ by Proposition 5.10. □

Thus ρ realizes V inside the corner $p_0 + s_1(\cdot)t_1$, acting as the identity on the complementary corner $s_0(\cdot)t_0$: this complementarity is the algebraic seed of the centralizer relation $[K, J] = 1$.

ThompsonVWitness
`thompsonV_not_isLEF`
ThompsonWitness
`not_isLEF_cornerWitnessSubgroup`
Scheme
`not_isLEF_cornerSubgroup`
Strengthened: proved for V itself, with no Higman presentation input.

Proposition 5.12. *Thompson's group V is finitely generated and is not LEF. Moreover V is perfect.*

Proof. V is finitely presented and infinite [11, 6] and even 2-generated [5]. If V were LEF, then being finitely presented it would be residually finite by Lemma 2.8; but V is infinite and simple, so its only finite quotient is trivial and it is not residually finite. Perfectness: $[V, V]$ is a nontrivial normal subgroup of the simple group V (it is nontrivial since V is nonabelian, containing for instance a copy of every finite symmetric group permuting the leaves of a fixed tree), hence $[V,V]=V$. ☐

Remark 5.13 (A two-relator route with a smaller trust surface)**.** The proof above spends three citations—infinitude, simplicity, finite presentability. There is a route that spends none, and it is the one the accompanying formalization takes; we record it because its trust surface is strictly smaller. Suppose two elements a, b of a group satisfy the two standard relators of Thompson's group F,

ThompsonFObstruction
`not_isLEF_of_two_relations`
`finite_commute_of_two_relations`
ThompsonWitness
`cornerWitnessSubgroup`
`not_isLEF_cornerWitnessSubgroup`
ThompsonVWitness
`thompsonV_not_isLEF`

$$[\,ab^{-1},\, a^{-1}ba\,]=1, \qquad [\,ab^{-1},\, a^{-2}ba^2\,]=1,$$

and do not commute. Then the group is not LEF. Indeed, in a *finite* group these two relations force $[a,b]=1$: setting $c_n=a^{-n}ba^n$, the relations propagate to $b^{-1}c_nb=c_{n+1}$ for all $n\geq 1$, and periodicity of $n\mapsto a^n$ closes an induction that drags $c_0=b$ into the centralizer of b's conjugates; a local embedding of a large enough finite subset would transfer the noncommuting pair, with its two relations, into a finite group. Now V contains two explicit tree tables satisfying the F-relators and not commuting (any embedded copy of F provides them), so V is not LEF—with no appeal to a presentation of V, to simplicity, or to Lemma 2.8. In the formalization the witness inside the corner is a 2-generated subgroup of explicit cylinder units satisfying the two relators, and its elementary membership is discharged by explicit cylinder-commutator and Whitehead identities rather than by perfectness of V; nothing in the closed proof identifies any group with V.

6. The compression scheme in adjacent ranks

This section presents the algebraic heart of the paper in its natural generality. The matrix identities below are verifiable by finite computation from the two Leavitt relations, over an arbitrary ambient ring. Beyond them, the section uses only the facts about Thompson's group V established in Section 5 and the two membership hypotheses isolated in Theorem 6.7, the latter discharged explicitly in Section 7 and K-theoretically in Section 8.

Throughout, A is a nontrivial associative unital ring carrying a binary Leavitt family (s_0,s_1,t_0,t_1) in the sense of Definition 5.1—in particular A may be, but need not contain a copy of, the Leavitt algebra $L=L_k(1,2)$—and we retain the notation $p_0=s_0t_0$, $p_1=s_1t_1$. Fix $m\geq 2$ and put

$$n=m+1, \qquad G=\mathrm{EL}_n(A), \qquad \Gamma=\{\hat\gamma=\mathrm{diag}(\gamma,1)\,|\,\gamma\in\mathrm{EL}_m(A)\}\ \leq\ G.$$

Indeed $\Gamma\leq G$: an elementary matrix in the first m coordinates is an elementary matrix of rank n. The group $\Gamma\cong\mathrm{EL}_m(A)$ is infinite, since $a\mapsto x_{12}(a)$ is injective and A is infinite by Lemma 5.2.

We use the left comb (22) with n leaves:

$$\alpha_i=s_0^{i-1}s_1 \quad (1\leq i\leq m), \qquad \alpha_n=s_0^m, \tag{24}$$

a complete leaf set, so that $\alpha_i^*\alpha_j=\delta_{ij}$ and $\sum_{i=1}^n\alpha_i\alpha_i^*=1$ by Lemma 5.3.

6.1. **The comb compressor.** Define the column $b \in A^{m\times 1}$, the row $c \in A^{1\times m}$, and the matrices $u, u' \in M_n(A)$:

$$b = \begin{pmatrix} s_1\alpha_1^* \\ \vdots \\ s_1\alpha_m^* \end{pmatrix}, \qquad c = \begin{pmatrix} \alpha_1 t_1 & \cdots & \alpha_m t_1 \end{pmatrix}, \tag{25}$$
$$u = \begin{pmatrix} s_0 I_m & b \\ 0 & \alpha_n^* \end{pmatrix}, \qquad u' = \begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix}.$$

LeavittMatrixCompression `matrixCompressionHom`

Lemma 6.1. *The following identities hold in A:*

$$bc = p_1 I_m, \quad cb = 1 - \alpha_n\alpha_n^*, \quad t_0 b = 0, \quad cs_0 = 0, \quad b\alpha_n = 0, \quad \alpha_n^* c = 0.$$

Consequently $uu' = u'u = I_n$; thus $u \in \mathrm{GL}_n(A)$ with $u^{-1} = u'$.

Proof. Entrywise: $(bc)_{ij} = s_1\alpha_i^*\alpha_j t_1 = \delta_{ij}s_1t_1 = \delta_{ij}p_1$. Next, $cb = \sum_{i=1}^m \alpha_i t_1 s_1 \alpha_i^* = \sum_{i=1}^m \alpha_i\alpha_i^* = 1 - \alpha_n\alpha_n^*$ by Lemma 5.3. The entries of $t_0 b$ are $t_0 s_1 \alpha_i^* = 0$; those of cs_0 are $\alpha_i t_1 s_0 = 0$; those of $b\alpha_n$ are $s_1\alpha_i^*\alpha_n = 0$ and of $\alpha_n^* c$ are $\alpha_n^*\alpha_i t_1 = 0$, by orthogonality of distinct leaves. Then

$$uu' = \begin{pmatrix} s_0t_0I_m + bc & b\alpha_n \\ \alpha_n^* c & \alpha_n^*\alpha_n \end{pmatrix} = \begin{pmatrix} (p_0+p_1)I_m & 0 \\ 0 & 1 \end{pmatrix} = I_n,$$

and likewise

$$u'u = \begin{pmatrix} t_0s_0I_m & t_0 b \\ cs_0 & cb + \alpha_n\alpha_n^* \end{pmatrix} = I_n. \qquad \square$$

RankFourCompressors `compressor_conjugation` `compressionEnd_injective`

Proposition 6.2 (Compression)**.** *For every $\gamma \in \mathrm{EL}_m(A)$,*

$$u\,\hat\gamma\,u^{-1} = p_1\widehat{I_m + s_0}\gamma t_0\,, \tag{26}$$

and on elementary generators $u\,\hat x_{ij}(a)\,u^{-1} = \hat x_{ij}(s_0at_0)$ for $i \neq j \leq m$. Consequently

$$K := u\Gamma u^{-1} = \langle\, \hat x_{ij}(s_0at_0) \;:\; a \in A,\ i \neq j \leq m \,\rangle \;\leq\; \Gamma, \tag{27}$$

and conjugation by u is an isomorphism $\Gamma \xrightarrow{\cong} K$; in particular $K \cong \mathrm{EL}_m(A)$.

Proof. Block multiplication using Lemma 6.1:

$$u\,\hat\gamma\,u^{-1} = \begin{pmatrix} s_0\gamma & b \\ 0 & \alpha_n^* \end{pmatrix}\begin{pmatrix} t_0 I_m & 0 \\ c & \alpha_n \end{pmatrix} = \begin{pmatrix} s_0\gamma t_0 + bc & b\alpha_n \\ \alpha_n^* c & \alpha_n^*\alpha_n \end{pmatrix}$$
$$= \begin{pmatrix} s_0\gamma t_0 + p_1 I_m & 0 \\ 0 & 1 \end{pmatrix}.$$

For $\gamma = x_{ij}(a) = I_m + aE_{ij}$, the block is $p_1I_m + s_0t_0I_m + s_0at_0E_{ij} = I_m + s_0at_0E_{ij} = x_{ij}(s_0at_0)$. Since Γ is generated by the $\hat x_{ij}(a)$ and conjugation is a group isomorphism onto its image, K is generated by the displayed elements, which lie in Γ. $\square$

The compressed copy K sits inside Γ with all matrix entries confined to the corner $s_0(\cdot)t_0 = p_0Ap_0$ of A; the complementary corner p_1Ap_1 is untouched, and this is the room in which the centralizing subgroup J will live.

6.2. **The involution z and generation.** Let $z \in M_n(A)$ agree with the identity matrix outside the coordinates 1 and n, where it is

(28)

$$z\big|_{\{1,n\}} = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}; \qquad \text{that is,} \qquad z = I_n + (p_0 - 1)E_{11} + s_1 E_{1n} + t_1 E_{n1} - E_{nn}.$$

Lemma 6.3. *$z^2 = I_n$; in particular $z \in \mathrm{GL}_n(A)$. Moreover z centralizes K.* Leavitt z_sq

Proof. On the coordinates $\{1, n\}$, RankFourCompressors involution_sq

$$\begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = \begin{pmatrix} p_0 + p_1 & 0 \\ 0 & 1 \end{pmatrix} = I_2,$$

DiagonalCornerCompression matrixCompression_commutes_firstDiagonalCorner

using $p_0 s_1 = 0$, $t_1 p_0 = 0$, $t_1 s_1 = 1$; the remaining coordinates are fixed. For the centralizing statement, put $r = s_0 a t_0$ and note

$$p_0 r = r = r p_0, \qquad t_1 r = 0, \qquad r s_1 = 0.$$

Let $i \neq j \leq m$. If $1 \notin \{i, j\}$, then z acts as the identity on the coordinates i, j and $z\, \hat{x}_{ij}(r)\, z = \hat{x}_{ij}(r)$. If $i = 1$: the matrix rE_{1j} satisfies $z(rE_{1j})z = (p_0 r E_{1j} + t_1 r E_{nj})z = rE_{1j}z = rE_{1j}$, since the j-th row of z is $e_j^\intercal$ for $2 \leq j \leq m$. If $j = 1$: $z(rE_{i1})z = rE_{i1}z = rp_0 E_{i1} + rs_1 E_{in} = rE_{i1}$. In each case $z\, \hat{x}_{ij}(r)\, z^{-1} = \hat{x}_{ij}(r)$, and these elements generate K by (27). □

Define the second compressor

$$v = zu. \tag{29}$$

Then, by Proposition 6.2 and Lemma 6.3,

$$v\Gamma v^{-1} = z(u\Gamma u^{-1})z^{-1} = zKz^{-1} = K = u\Gamma u^{-1} : \tag{30}$$

the two compressors have one common image. The quotient $vu^{-1} = z$ is precisely what generates the missing last row and column:

Proposition 6.4 (Generation)**.** *The subgroup of $\mathrm{GL}_n(A)$ generated by Γ and z contains $\mathrm{EL}_n(A)$. Consequently, if u and z lie in $\mathrm{EL}_n(A)$, then* RankFourCompressors coreEmbedding_compressorSet_generate

$$\mathrm{EL}_n(A) = \langle \Gamma, z \rangle = \langle \Gamma, u, v \rangle.$$

Proof. For $2 \leq j \leq m$ and $a \in A$, we claim

$$z\, \hat{x}_{1j}(s_1 a)\, z = x_{nj}(a), \qquad z\, \hat{x}_{j1}(a t_1)\, z = x_{jn}(a).$$

For the first: $z(s_1 a E_{1j})z = (p_0 s_1 a E_{1j} + t_1 s_1 a E_{nj})z = aE_{nj}z = aE_{nj}$, using $p_0 s_1 = 0$, $t_1 s_1 = 1$, and that the j-th row of z is $e_j^\intercal$. For the second: $z(a t_1 E_{j1})z = a t_1 E_{j1} z = a t_1 p_0 E_{j1} + a t_1 s_1 E_{jn} = aE_{jn}$, using $t_1 p_0 = 0$, $t_1 s_1 = 1$, and that the j-th column of z is e_j. Thus $\langle \Gamma, z \rangle$ contains $x_{nj}(a)$ and $x_{jn}(a)$ for all $2 \leq j \leq m$ and $a \in A$, as well as all $\hat{x}_{ij}(a) = x_{ij}(a)$ with $i \neq j \leq m$. The remaining elementary matrices are supplied by the Steinberg relation, using $m \geq 2$ so that the coordinate $2 \notin \{1, n\}$ exists:

$$x_{1n}(a) = [\, x_{12}(a),\, x_{2n}(1)\,], \qquad x_{n1}(a) = [\, x_{n2}(a),\, x_{21}(1)\,].$$

Hence $\langle \Gamma, z\rangle \supseteq \mathrm{EL}_n(A)$. If moreover $u, z \in \mathrm{EL}_n(A)$, then $\langle \Gamma, z\rangle \subseteq \mathrm{EL}_n(A)$, giving the first equality; and since $z = vu^{-1}$, also $\langle \Gamma, u, v\rangle \supseteq \langle \Gamma, z\rangle = \mathrm{EL}_n(A) \supseteq \langle \Gamma, u, v\rangle$. □

6.3. **The commuting corner copy of V.** For $g \in V$, recall $\rho(g) = p_0 + s_1 U_g t_1 \in A^\times$ from Lemma 5.11, and define

$$j(g) = \mathrm{diag}\big(\rho(g), 1, \ldots, 1\big) \in \mathrm{GL}_n(A), \qquad J = \{j(g) \,|\, g \in V\}.$$

UniversalRankFour
`witnessEmbedding`
DiagonalElementary
`firstDiagonalUnit_mem_of_`
`mem_commutator`

Proposition 6.5. *For every $g \in V$, the matrix* $\mathrm{diag}(\rho(g), 1, \ldots, 1) \in \mathrm{GL}_m(A)$ *lies in* $\mathrm{EL}_m(A)$. *Hence $J \leq \Gamma$, the map $j : V \to J$ is an isomorphism, and J is a finitely generated subgroup of Γ that is not LEF.*

Proof. By Proposition 5.12, V is perfect, so $g = \prod_i [a_i, b_i]$ for some $a_i, b_i \in V$. Since ρ is a homomorphism (Lemma 5.11), $\rho(g) = \prod_i [\rho(a_i), \rho(b_i)]$ is a product of commutators of units of A. The Whitehead commutator lemma (Lemma 5.5, with $m \geq 2$) places $\mathrm{diag}(\rho(g), 1, \ldots, 1)$ in $\mathrm{EL}_m(A)$. Thus $j(g) \in \Gamma$. The map j is a homomorphism because ρ is, and it is injective because ρ is; so $J \cong V$, which is finitely generated and not LEF by Proposition 5.12. □

Proposition 6.5 holds for every unital ring A containing the Leavitt family: the corner copy of V is elementary with no hypothesis on A beyond the Leavitt family itself.

Scheme
`commute_compressed_corner`
`compressed_inf_corner`

Proposition 6.6. $[K, J] = 1$ *and* $K \cap J = \{1\}$. *Consequently* $KJ \cong K \times J$.

Proof. Centralizing. Conjugating an elementary matrix by a diagonal matrix multiplies its coefficient: $j(g)\,\hat{x}_{il}(r)\,j(g)^{-1} = \hat{x}_{il}\big(\rho(g)^{[i=1]}\,r\,\rho(g^{-1})^{[l=1]}\big)$, where the exponent $[i = 1]$ is 1 if $i = 1$ and 0 otherwise. For $r = s_0 a t_0$, Lemma 5.11 gives $\rho(g) r = \rho(g) s_0\, a t_0 = s_0 a t_0 = r$ and $r \rho(g^{-1}) = s_0 a\, t_0 \rho(g^{-1}) = s_0 a t_0 = r$. Hence $j(g)$ commutes with every generator $\hat{x}_{il}(s_0 a t_0)$ of K from (27).

Trivial intersection. Suppose $j(g) \in K$. By (26), every element of K has the form $\widehat{p_1 I_m + s_0 \gamma t_0}$ with $\gamma \in \mathrm{EL}_m(A)$; comparing $(1,1)$-entries, $\rho(g) = p_1 + s_0 \gamma_{11} t_0$. Sandwiching between t_1 and s_1: on one side, $t_1(\rho(g) - 1)s_1 = t_1(s_1 U_g t_1 - p_1)s_1 = U_g - 1$, using $t_1 p_0 = 0$; on the other, $t_1(p_1 + s_0\gamma_{11} t_0 - 1)s_1 = t_1(s_0 \gamma_{11} t_0 - p_0)s_1 = 0$, using $t_1 s_0 = 0$ and $t_1 p_0 = 0$. Hence $U_g = 1$, so $g = 1$ by Proposition 5.10, and $j(g) = 1$.

Since K and J commute and intersect trivially, the multiplication map $K \times J \to KJ$ is an isomorphism. □

6.4. **The Leavitt-corner theorem.**

CompressionSetup
`CompressionSetup`
UniversalCompressionSetup
`compressionSetup`

Theorem 6.7 (Leavitt-corner theorem)**.** *Let A be a nontrivial associative unital ring carrying a binary Leavitt family, let $m \geq 2$, and put $n = m + 1$. Let $u, z \in \mathrm{GL}_n(A)$ be the matrices* (25) *and* (28)*. Assume:*

(i) $\mathrm{EL}_m(A)$ *and* $\mathrm{EL}_n(A)$ *have property* (T)*;*
(ii) $u, z \in \mathrm{EL}_n(A)$*; this holds, for instance, whenever* $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$.

Then $\mathrm{EL}_n(A)$ *is not sofic.*

Proof. Put $G = \mathrm{EL}_n(A)$ and let Γ, u, z, $v = zu$, K, J be as above. By hypothesis (i), G and $\Gamma \cong \mathrm{EL}_m(A)$ are property-(T) groups, hence finitely generated, and Γ is infinite. By hypothesis (ii) and Proposition 6.4, $G = \langle \Gamma, u, v\rangle = \langle \Gamma, Q\rangle$ with $Q = \{u, v\} \subseteq G$. By Proposition 6.2 and (30),

$$u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma,$$

so both elements of Q conjugate Γ into itself. Fix $q_0 = u$, so $K = q_0\Gamma q_0^{-1}$. By Propositions 6.5 and 6.6, the subgroup $J \leq \Gamma$ is finitely generated with $[K, J] = 1$ and $K \cap J = \{1\}$, and $J \cong V$ is not LEF. If G were sofic, Corollary 4.2 (Theorem D) would force J to be LEF, a contradiction. □

Corollary 6.8. *Let A be a nontrivial* finitely generated *unital ring carrying a binary Leavitt family, and let $m \geq 3$, $n = m + 1$. If $u, z \in \mathrm{EL}_n(A)$—for instance, if $\mathrm{GL}_n(A) = \mathrm{EL}_n(A)$—then $\mathrm{EL}_n(A)$ is an infinite, finitely generated, property-(T) group that is not sofic.* *CriterionAssembly* `not_isSofic_of_not_isLEF`

Proof. Theorem 5.7 supplies hypothesis (i) of Theorem 6.7 for both ranks $m, n \geq 3$, so $\mathrm{EL}_n(A)$ is not sofic and has property (T). It is finitely generated by Lemma 5.6, and infinite since it contains $J \cong V$. □

Remark 6.9 (The trust surface of the scheme)**.** The matrix identities of this section are verifiable by finite computation from the two Leavitt relations; beyond them, the section inputs only the Whitehead lemma, the finite generation, perfectness, and non-LEF property of V established in Section 5, and hypotheses (i) and (ii) of Theorem 6.7. The corner copy of V is elementary for every ambient ring: Proposition 6.5 uses only the perfectness of V and the Whitehead lemma. The entire membership burden therefore falls on hypothesis (ii), which concerns exactly two matrices, u and z: Section 7 discharges it by explicit elementary factorizations in characteristic two, and Section 8 discharges it over every finite field via $K_1(L) = 0$ and the GE property, with the K_1 input available in elementary, formalized form (Remark 8.5).

7. The explicit realization: no K-theory

In this section $k = \mathbb{F}_2$ and $R = L_{\mathbb{F}_2}(1, 2)$; the matrix identities and the argument extend without change to any finite field of characteristic two. Fix $m \geq 3$, put $n = m + 1$, and let $G = \mathrm{EL}_n(R)$, Γ, u, z, v, K, J be as in Section 6 with $A = R$. We now exhibit the two matrices u and z as explicit products of elementary matrices, discharging hypothesis (ii) of Theorem 6.7 with no K-theoretic input.

Proposition 7.1 (Explicit elementary factorization of the compressor)**.** *For $1 \leq i \leq m$, put* *RankFourCompressors* `compressor` `compressorMatrix` `compressorPiece` *Explicit elementary words, in every characteristic.*

$$U_i = x_{ni}(1 + t_0)\, x_{in}(1)\, x_{ni}(1 + s_0)\, x_{in}(t_0) \;\in\; \mathrm{EL}_n(R).$$

Then

$$u = U_m\, U_{m-1} \cdots U_1.$$

In particular $u \in \mathrm{EL}_n(R)$, as a product of $4m$ elementary matrices.

Proof. By Lemma 5.9(b), applied in the pair of coordinates (i,n), the matrix U_i agrees with the identity outside the coordinates i and n and equals $\left(\begin{smallmatrix} s_0 & s_1t_1 \\ 0 & t_0 \end{smallmatrix}\right)$ on them; explicitly,

$$U_i = I_n + (s_0+1)E_{ii} + s_1t_1E_{in} + (t_0+1)E_{nn},$$

signs being immaterial in characteristic two. Put $P_k = U_kU_{k-1}\cdots U_1$. We claim, by induction on k, that

$$P_k = \sum_{i\le k}(s_0E_{ii} + s_1t_1t_0^{i-1}E_{in}) + \sum_{k<i\le m} E_{ii} + t_0^kE_{nn}.$$

For $k=1$ this is the displayed form of U_1. For the inductive step, compute the rows of $P_{k+1} = U_{k+1}P_k$. For $i \notin \{k+1, n\}$, the i-th row of U_{k+1} is e_i^T, so the i-th row of P_{k+1} equals that of P_k. The $(k+1)$-st row of U_{k+1} is $s_0e_{k+1}^\mathsf{T} + s_1t_1e_n^\mathsf{T}$; since the $(k+1)$-st row of P_k is e_{k+1}^T and the n-th row of P_k is $t_0^ke_n^\mathsf{T}$, the $(k+1)$-st row of P_{k+1} is $s_0e_{k+1}^\mathsf{T} + s_1t_1t_0^ke_n^\mathsf{T}$. The n-th row of U_{k+1} is $t_0e_n^\mathsf{T}$, so the n-th row of P_{k+1} is $t_0^{k+1}e_n^\mathsf{T}$. This is the claimed form for $k+1$.

Finally, for the left comb (24), $s_1t_1t_0^{i-1} = s_1(s_0^{i-1}s_1)^* = s_1\alpha_i^*$ for $1\le i\le m$, and $t_0^m = \alpha_n^*$. Hence

$$P_m = \sum_{i=1}^m (s_0E_{ii} + s_1\alpha_i^*E_{in}) + \alpha_n^*E_{nn} = \begin{pmatrix} s_0I_m & b \\ 0 & \alpha_n^* \end{pmatrix} = u,$$

by (25). ☐

RankFourCompressors
`involution`
`involutionMatrix`
`involution_sq`
Explicit elementary words, in every characteristic.

Lemma 7.2 (Explicit elementary factorization of the involution)**.**

$$z = x_{1n}(s_1)\, x_{n1}(t_1)\, x_{1n}(s_1) \;\in\; \mathrm{EL}_n(R).$$

Proof. Lemma 5.9(a), applied in the pair of coordinates $(1,n)$, produces exactly the matrix z of (28). ☐

MainResults
`universalLeavittEL4_not_isSofic`
`universalLeavitt_profile`

Theorem 7.3 (The explicit spine)**.** *For every $m \ge 3$, the group $\mathrm{EL}_{m+1}(R)$ over $R = L_{\mathbb{F}_2}(1,2)$ is infinite, finitely generated, has property (T), and is not sofic.*

Proof. The ring $R = L_{\mathbb{F}_2}(1,2)$ is finitely generated (by s_0, s_1, t_0, t_1 over the prime field). Proposition 7.1 and Lemma 7.2 verify $u, z \in \mathrm{EL}_{m+1}(R)$, and Corollary 6.8 applies. ☐

Proof of Theorem A. Theorem 7.3 gives every $m\ge 3$; the case $m=3$ is the group $G = \mathrm{EL}_4(R)$. For $m\in\{1,2\}$, the elementary rank equivalence of Proposition 5.8 is an explicit isomorphism $\mathrm{EL}_{m+1}(R) \cong \mathrm{EL}_4(R)$, and all four properties transport along it; in particular $\mathrm{EL}_2(R)$ and $\mathrm{EL}_3(R)$ have property (T) although Theorem 5.7 says nothing at rank two. The two auxiliary matrices are exhibited as explicit products of elementary matrices: u by Proposition 7.1 and z (hence $v = zu$) by Lemma 7.2. The remaining memberships, of the diagonal matrices $j(g)$, $g\in V$, follow from the explicit Whitehead identity of Lemma 5.5 together with the perfectness of V (Proposition 6.5); once a commutator expression of g in V is chosen, the Whitehead identity converts it into an explicit elementary word. No K-theory is used. □

Remark 7.4 (Scope of the explicit spine)**.** The factorizations of this section are identities in characteristic two and hold over any field of characteristic two; finiteness of the field is used only through finite generation of the ring in Theorem 5.7. Thus $\mathrm{EL}_{m+1}(L_k(1,2))$ is infinite, finitely generated, Kazhdan, and nonsofic, with all memberships explicit, for every finite field k of characteristic two and every $m \geq 3$—and then for every $m \geq 1$ by the elementary rank equivalence (Proposition 5.8), still with no K-theory. For rank $n = 3$ (that is, $m = 2$), hypothesis (i) of Theorem 6.7 for EL_2 is not supplied by Theorem 5.7; it is exactly here that the rank equivalence substitutes. Section 8 reaches the remaining groups of Theorem B—the unit group and every GL_r, over every finite field—through self-similarity.

8. Rank two and the full panorama

The compression–centralizer mechanism closes already in rank two, over a different and smaller group Γ: the full diagonal unit group, with property (T) transported through self-similarity rather than supplied by Theorem 5.7. Since all the groups in question turn out to be isomorphic, the value of the rank-two realization is not a new group but a map of the mechanism's boundary—it shows how little room the criterion actually needs. Combined with matrix self-similarity and the vanishing of K_1 (Proposition 8.3, with the elementary proof of Appendix A), this yields Theorem B.

8.1. The two-by-two realization.

Theorem 8.1 (Two-by-two Leavitt-corner theorem)**.** *Let A be a nontrivial associative unital ring carrying a binary Leavitt family. Suppose that $A^\times$ and $\mathrm{EL}_2(A)$ have property (T) and that $\mathrm{GL}_2(A) = \mathrm{EL}_2(A)$. Then $\mathrm{EL}_2(A)$ is not sofic.*

RankTwoCompression
`rankTwo_not_isSofic`
`rankTwoInvolution_conj_upperNil`
`rankTwoInvolution_conj_lowerNil`

Proof. Put $G = \mathrm{EL}_2(A) = \mathrm{GL}_2(A)$ and

$$\Gamma = \{\mathrm{diag}(a,1) \,|\, a \in A^\times\} \leq G.$$

Then $\Gamma \cong A^\times$ is a property-(T) group, hence finitely generated, and it is infinite because $\rho : V \hookrightarrow A^\times$ is injective (Lemma 5.11).

The compressor. Put

$$u = \begin{pmatrix} s_0 & p_1 \\ 0 & t_0 \end{pmatrix}, \qquad u' = \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix}.$$

Using $p_1 s_0 = s_1(t_1 s_0) = 0$, $t_0 p_1 = (t_0 s_1)t_1 = 0$, $t_0 s_0 = 1$, and $p_0 + p_1 = 1$:

$$uu' = \begin{pmatrix} s_0 t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = I_2, \qquad u'u = \begin{pmatrix} t_0 s_0 & t_0 p_1 \\ p_1 s_0 & p_1 + s_0 t_0 \end{pmatrix} = I_2,$$

so $u \in \mathrm{GL}_2(A) = G$ with $u^{-1} = u'$. For $a \in A$, put

$$\kappa(a) = p_1 + s_0 a t_0.$$

Then $\kappa(1) = 1$ and $\kappa(a)\kappa(a') = p_1 + s_0 a a' t_0 = \kappa(aa')$ by the same identities, so κ is unital and multiplicative and restricts to a group homomorphism $A^\times \to A^\times$, injective because $t_0 \kappa(a) s_0 = t_0 p_1 s_0 + t_0 s_0 a t_0 s_0 = a$; and

$$\begin{aligned} u\, \mathrm{diag}(a,1)\, u^{-1} &= \begin{pmatrix} s_0 a & p_1 \\ 0 & t_0 \end{pmatrix} \begin{pmatrix} t_0 & 0 \\ p_1 & s_0 \end{pmatrix} \\ &= \begin{pmatrix} s_0 a t_0 + p_1 & p_1 s_0 \\ t_0 p_1 & t_0 s_0 \end{pmatrix} = \mathrm{diag}(\kappa(a), 1). \end{aligned}$$

Hence

$$K := u \Gamma u^{-1} = \{\mathrm{diag}(\kappa(a),1) \,|\, a \in A^\times\} \le \Gamma.$$

The involution. Put

$$z = \begin{pmatrix} p_0 & s_1 \\ t_1 & 0 \end{pmatrix}, \qquad \text{so} \qquad z^2 = \begin{pmatrix} p_0^2 + s_1 t_1 & p_0 s_1 \\ t_1 p_0 & t_1 s_1 \end{pmatrix} = I_2,$$

using $p_0 s_1 = 0$, $t_1 p_0 = 0$, $t_1 s_1 = 1$; thus $z \in \mathrm{GL}_2(A) = G$ and $z^{-1} = z$. For any $c \in A$,

$$z\, \mathrm{diag}(c,1)\, z = \begin{pmatrix} p_0 c p_0 + p_1 & p_0 c s_1 \\ t_1 c p_0 & t_1 c s_1 \end{pmatrix}. \tag{31}$$

The element $\kappa(a)$ satisfies $p_0 \kappa(a) = \kappa(a) p_0 = s_0 a t_0$, $t_1 \kappa(a) = t_1$, and $\kappa(a) s_1 = s_1$; substituting into (31) gives $z\, \mathrm{diag}(\kappa(a),1)\, z = \mathrm{diag}(s_0 a t_0 + p_1, 1) = \mathrm{diag}(\kappa(a),1)$. Hence z centralizes K, and $v := zu \in G$ satisfies $v \Gamma v^{-1} = z K z^{-1} = K$.

Generation. For any $a \in A$, the elements $1 + p_0 a t_1$ and $1 + s_1 a p_0$ are units of A, since $(p_0 a t_1)^2 = 0$ and $(s_1 a p_0)^2 = 0$ (using $t_1 p_0 = 0$ and $p_0 s_1 = 0$); so the diagonal matrices $\mathrm{diag}(1 + p_0 a t_1, 1)$ and $\mathrm{diag}(1 + s_1 a p_0, 1)$ lie in Γ. Substituting $c = 1 + p_0 a t_1$ and $c = 1 + s_1 a p_0$ into (31) and using $t_1 p_0 = 0$, $p_0 s_1 = 0$, $t_1 s_1 = 1$, $p_0^2 = p_0$:

$$z\, \mathrm{diag}(1 + p_0 a t_1,\, 1)\, z = \begin{pmatrix} 1 & p_0 a \\ 0 & 1 \end{pmatrix} = x_{12}(p_0 a), \tag{32}$$

$$z\, \mathrm{diag}(1 + s_1 a p_0,\, 1)\, z = \begin{pmatrix} 1 & 0 \\ a p_0 & 1 \end{pmatrix} = x_{21}(a p_0). \tag{33}$$

Next, the *corner swap*

$$d = s_1 t_0 + s_0 t_1 \in A^\times, \qquad d^2 = s_1 t_1 + s_0 t_0 = 1,$$

gives $D = \mathrm{diag}(d,1) \in \Gamma$ with $D^{-1} = D$. Conjugation by D multiplies an upper coefficient on the left by d and a lower coefficient on the right by d; since $d\, p_0 s_0 t_1 = d\, s_0 t_1 = p_1$ and $s_1 t_0 p_0\, d = s_1 t_0\, d = p_1$,

$$D\, x_{12}(p_0(s_0 t_1 a))\, D = x_{12}(p_1 a), \qquad D\, x_{21}((a s_1 t_0) p_0)\, D = x_{21}(a p_1),$$

with the inner matrices supplied by (32) and (33). Finally

$$x_{12}(a) = x_{12}(p_0 a)\, x_{12}(p_1 a), \qquad x_{21}(a) = x_{21}(a p_0)\, x_{21}(a p_1),$$

so $\langle \Gamma, z\rangle$ contains every elementary matrix; hence $\langle \Gamma, z\rangle = G$, and since $z = vu^{-1}$,

$$G = \langle \Gamma, u, v\rangle, \qquad u\Gamma u^{-1} = v\Gamma v^{-1} = K \leq \Gamma.$$

The commuting corner. Put $J = \{\operatorname{diag}(\rho(g), 1) \mid g \in V\} \leq \Gamma$; membership is automatic here, since $\rho(g) \in A^\times$. By Lemma 5.11 and Proposition 5.12, $J \cong V$ is finitely generated and not LEF. The subgroups K and J commute:

$$\begin{aligned} \kappa(a)\rho(g) &= (p_1 + s_0 a t_0)(p_0 + s_1 U_g t_1) = s_1 U_g t_1 + s_0 a t_0 \\ &= (p_0 + s_1 U_g t_1)(p_1 + s_0 a t_0) = \rho(g)\kappa(a), \end{aligned}$$

using $p_1 s_1 = s_1$, $t_1 p_1 = t_1$, $t_0 p_0 = t_0$, $p_0 s_0 = s_0$, $t_0 s_1 = 0$, $t_1 s_0 = 0$. And $K \cap J$ is trivial: if $\kappa(a) = \rho(g)$, then sandwiching between t_0 and s_0 gives $a = t_0 \kappa(a) s_0 = t_0 \rho(g) s_0 = 1$, so $\rho(g) = \kappa(1) = 1$ and $g = 1$ by injectivity of ρ.

Conclusion. The data G, Γ, $Q = \{u, v\}$, $q_0 = u$, K, J satisfy all hypotheses of Corollary 4.2: G and Γ are finitely generated Kazhdan groups, Γ is infinite, $G = \langle \Gamma, Q\rangle$, each $q \in Q$ conjugates Γ into itself with $q_0 \Gamma q_0^{-1} = K$, and $J \leq \Gamma$ is finitely generated with $[K, J] = 1$ and $K \cap J = \{1\}$. If G were sofic, $J \cong V$ would be LEF, contradicting Proposition 5.12. ☐

8.2. **K_1 vanishes and $\mathrm{GL} = \mathrm{EL}$.** We isolate the ring-theoretic external input in the exact form used.

Theorem 8.2 (Ara–Goodearl–Pardo [3])**.** *Let R be a purely infinite simple unital ring. Then:*

(a) *R is a GE-ring: for every $r \geq 1$, every matrix in $\mathrm{GL}_r(R)$ is a product of elementary matrices and a matrix $\operatorname{diag}(a, 1, \ldots, 1)$ with $a \in R^\times$;*
(b) *the natural map $R^\times = \mathrm{GL}_1(R) \to K_1(R)$ is surjective and induces an isomorphism $R^\times/[R^\times, R^\times] \cong K_1(R)$.*

formalized in part
MatrixDiagonalization
`exists_elementary_mul_diag`
`binaryLeavitt_exists_`
`elementary_mul_diag`
LeavittSimplicity
`exists_mul_mul_eq_one`
The rank-two GE form is Lean-backed; part (b) remains a cited input.

Proposition 8.3. *Let k be any field and $L = L_k(1, 2)$. Then:*

(i) *$K_1(L) = 0$;*
(ii) *the unit group $L^\times$ is perfect;*
(iii) *$\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for every $r \geq 2$.*

formalized in part
RefineLoopDischarge
`glTwo_eq_elementary_holds`
`glFour_eq_elementary_holds`
`stableUnits_eq_top_holds`
$K_1(L) = 0$ in Whitehead form is unconditional; the $\mathrm{GL} = \mathrm{EL}$ collapse is formalized at ranks two and four.

Proof. Parts (i) and (iii) are proved, elementarily and in full, in Appendix A: Theorem A.2 establishes $K_1(L) = 0$ in Whitehead form—every unit u has $\operatorname{diag}(u, 1) \in \mathrm{EL}_2(L)$—and Corollary A.23 deduces $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for every $r \geq 2$ via the elementary diagonalization of Lemma A.4. The only input of the appendix beyond the Leavitt relations is strong division (Lemma A.3), which holds because L is purely infinite simple: the graph criteria of [1] are immediate for the rose R_2 with one vertex and two loops—the single vertex admits no nontrivial hereditary saturated subset, every cycle has an exit, and the vertex lies on a cycle.

For (ii): Theorem 8.2(b) identifies $K_1(L)$ with $L^\times/[L^\times, L^\times]$, so (i) gives $L^\times = [L^\times, L^\times]$.

For an independent confirmation of (i), the K-theory of Leavitt path algebras [2] gives the exact sequence

$$K_1(k) \xrightarrow{1-N^{\mathsf{T}}} K_1(k) \longrightarrow K_1(L) \longrightarrow K_0(k) \xrightarrow{1-N^{\mathsf{T}}} K_0(k),$$

where N is the adjacency matrix—here the 1×1 matrix (2), so $1 - N^{\mathsf{T}} = -1$, an isomorphism on both $K_0(k) \cong \mathbb{Z}$ and $K_1(k) \cong k^{\times}$, whence $K_1(L) = 0$. $\square$

Remark 8.4. For an independent recent account of generation phenomena for unit groups and matrix groups over Leavitt algebras, consistent with Proposition 8.3 and with the self-similarity of Proposition 5.4, see [12].

RefineLoopDischarge
`K1_trivial`
`narrowReduction_holds`
`scalarReduction_holds`

Remark 8.5 (An elementary route to $K_1(L) = 0$)**.** The classical computation of $K_1(L)$ goes through the localization sequence of [2]. Appendix A replaces this input by a complete elementary proof—a Gaussian elimination over the binary tree, in the spirit of a Wiener–Hopf factorization whose "symbol" is a matrix pencil with two shift variables. In outline: a width reduction narrows every unit, modulo the Whitehead class group H of Definition A.1, to one whose value lies in the degree window $[-1, 1]$; at a sufficiently deep leaf set such a narrow unit becomes a matrix pencil with scalar coefficient matrices (A_0, A_1, C, B_0, B_1) over the five-dimensional space $kt_0 \oplus kt_1 \oplus k \oplus ks_0 \oplus ks_1$, indexed by a pair of complete leaf sets; and a single loop with two exits eliminates the pencil, splitting one column per pass (the index shift of the Wiener–Hopf analogy) until either a reshape by two tree tables lands the value in a one-sided degree window, or the stacked matrix $[B_0; B_1]$ acquires a scalar left inverse and the *inverse* unit lands there instead; one-sided windows die by a block-move induction whose base is the keystone Lemma A.16. Every step is carried out in Appendix A and formalized in the accompanying Lean development, so the K_1-theoretic input of [2] is eliminated from the trust surface of Theorem B; the citation is retained as an independent confirmation.

formalized in part
RefineLoopDischarge
`narrowReduction_holds`
The elimination behind it is formalized; the complexity bounds are manuscript-only.

Remark 8.6 (Effective Wiener–Hopf factorization over the binary tree)**.** The elimination of Remark 8.5, written out in Appendix A, is an algorithm, and it yields a factorization theorem of independent interest. Call a unit *nonpositive* (resp. *nonnegative*) if its value lies in the span of monomials $s_\alpha t_\beta$ with $|\alpha| \leq |\beta|$ (resp. $\geq$); these are the two "triangular" classes of the tree-coded Wiener–Hopf analogy, and each lies in H by an explicit block-move induction. The algorithm shows: *every unit of $L = L_k(1, 2)$ factors as a central scalar times a product of tree-table units (Thompson's group V), code-scalar units (elements of $\mathrm{GL}_n(k)$ transported along a complete prefix code), and a single nonpositive unit conjugated by two tree tables.* All factors are effectively computable: given u supported on monomials of depth at most d, the width reduction and the pencil loop run at codes with at most 2^{d+2} words, each loop step is Gaussian elimination over k on matrices of that size, and the loop executes at most 2^{d+1} times—so the factorization, and with it an explicit factorization of any $\mathrm{diag}(u, 1, \ldots, 1)$ into elementary matrices, is computed in time polynomial in 2^d and in the number of monomials of u. In the classical Wiener–Hopf picture the loop plays the role of index normalization: each refinement is the index shift that trades a constant diagonal for a shift atom, and the two exits are the two triangular factorizations. We are not aware of a prior effective factorization—or of any elementary proof of $K_1(L_k(1,2)) = 0$—in the literature; the localization-sequence computation of [2] is neither elementary nor effective.

8.3. Finite fields: all ranks and the unit group.

Theorem 8.7. *Let k be a finite field and $L = L_k(1,2)$. Then $\mathrm{EL}_2(L) = \mathrm{GL}_2(L)$ is infinite, finitely generated, has property (T), and is not sofic.*

RankTwoCompression
`rankTwo_not_isSofic`

Proof. The ring L is finitely generated, by s_0, s_1, t_0, t_1 and the finitely many scalars. By Theorem 5.7 and Lemma 5.6, $\mathrm{EL}_4(L)$ is a finitely generated Kazhdan group; by Proposition 8.3(iii), $\mathrm{EL}_4(L) = \mathrm{GL}_4(L)$; and by Proposition 5.4 with the left comb $\mathcal{C}_4$, $\mathrm{GL}_4(L) \cong L^\times$. Hence $L^\times$ is a finitely generated Kazhdan group. Similarly, $\mathrm{GL}_2(L) = \mathrm{EL}_2(L)$ by Proposition 8.3(iii), and $\mathrm{GL}_2(L) \cong L^\times$ by Proposition 5.4 with $\mathcal{C}_2 = (s_1, s_0)$; so $\mathrm{EL}_2(L)$ is Kazhdan as well. Theorem 8.1 with $A = L$ now shows that $\mathrm{EL}_2(L)$ is not sofic. It is infinite, containing $\mathrm{diag}(\rho(V), 1) \cong V$, and finitely generated, being Kazhdan. □

Theorem 8.8 (Panorama)**.** *Let k be a finite field and $L = L_k(1,2)$. Then the groups*

$$L^\times = \mathrm{GL}_1(L), \qquad \mathrm{GL}_r(L)\ (r \geq 1), \qquad \mathrm{EL}_r(L)\ (r \geq 2)$$

are mutually isomorphic—explicitly, $\Theta_{\mathcal{C}} : \mathrm{GL}_r(L) \to L^\times$ is an isomorphism for every ordered complete leaf set $\mathcal{C}$ of size r, and $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for $r \geq 2$—and each is an infinite, finitely generated, property-(T) group that is not sofic.

MainResults
`binaryLeavitt_finiteField_profile`
`binaryLeavittUnits_not_isSofic`
`binaryLeavittGL_not_isSofic`
`universalLeavittEL3_not_isSofic`
Proved over every finite field via self-similarity, without K_1 or GE.

Proof. The isomorphisms are Proposition 5.4, and the equalities $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ are Proposition 8.3(iii). Each of the listed groups is therefore isomorphic to $\mathrm{EL}_2(L)$, which is infinite, finitely generated, Kazhdan, and nonsofic by Theorem 8.7; all four properties are isomorphism-invariant. □

Proof of Theorem B. Part (i) and part (ii) are exactly Theorem 8.8, with the explicit isomorphism (2) given by Proposition 5.4. □

Remark 8.9 (Two routes to rank four)**.** For a finite field of characteristic two and rank $m + 1 \geq 4$, the membership hypotheses are discharged twice over: explicitly, with no K-theory, in Section 7; and via Proposition 8.3 here. Theorem B in its full strength—all ranks down to two, all finite fields, and the unit group itself—uses the second route. The step that makes the panorama characteristic-free is Proposition 6.5: the corner copy of V is elementary because its memberships go through perfectness and the Whitehead lemma, identities valid over every ring, rather than through transvection factorizations of transpositions, which require characteristic two.

9. Finitely presented nonsofic covers

Nonsoficity of a finitely generated group is witnessed by a finite fragment of its multiplication table; imposing that fragment as a presentation produces finitely presented nonsofic covers. The covers are proved nonsofic directly, from the forbidden table itself; quotient closure of soficity, which is unknown, plays no role.

TableCover `TableModel`

Definition 9.1. Let G be a group, $F \subseteq G$ finite with $1 \in F$, and $\varepsilon > 0$. An (F, ε)-*model* of G is a map $\varphi : F \cup F{\cdot}F \to \mathrm{Sym}(Y)$, for some finite set Y, such that

$$d_{\mathrm{H}}\big(\varphi(g)\varphi(h),\, \varphi(gh)\big) \le \varepsilon \qquad (g, h \in F), \tag{34}$$

$$d_{\mathrm{H}}\big(\varphi(g),\, \varphi(h)\big) \ge 1 - \varepsilon \qquad (g, h \in F,\ g \ne h). \tag{35}$$

TableCover `tableModel_of_isSofic` `exists_table_obstruction`

Lemma 9.2 (Finite-table characterization)**.** *A countable group G is sofic if and only if it admits an (F, ε)-model for every finite $F \ni 1$ and every $\varepsilon > 0$.*

Proof. If σ_n is a sofic approximation, then for fixed (F, ε) the restriction of σ_n to $F \cup F{\cdot}F$ is an (F, ε)-model for all large n: (34) is Definition 2.1(a), and (35) follows from (a) and (b) since $d_{\mathrm{H}}(\sigma_n(g), \sigma_n(h)) \ge d_{\mathrm{H}}(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(h)) - d_{\mathrm{H}}(\sigma_n(gh^{-1})\sigma_n(h), \sigma_n(g)) = d_{\mathrm{H}}(\sigma_n(gh^{-1}), \mathrm{id}) - o(1)$.

Conversely, exhaust G by finite symmetric sets $F_1 \subseteq F_2 \subseteq \cdots$ containing 1, and let φ_n be an $(F_n, 1/n)$-model on Y_n; replacing Y_n by many disjoint copies of itself we may assume $|Y_n| \to \infty$. Define $\sigma_n(g) = \varphi_n(g)$ for $g \in F_n$ and $\sigma_n(g) = \mathrm{id}$ otherwise. Fix $g, h \in G$; for n large, $g, h, gh \in F_n$, and (34) gives Definition 2.1(a). For faithfulness, first note that $p_n = \varphi_n(1)$ satisfies $d_{\mathrm{H}}(p_n^2, p_n) \le 1/n$ by (34) with $g = h = 1$, and $d_{\mathrm{H}}(p_n^2, p_n) = d_{\mathrm{H}}(p_n, \mathrm{id})$ by right invariance of d_{H}; so $d_{\mathrm{H}}(\varphi_n(1), \mathrm{id}) \le 1/n$. Then for $g \ne 1$ and n large,

$$d_{\mathrm{H}}(\sigma_n(g), \mathrm{id}) \ge d_{\mathrm{H}}(\varphi_n(g), \varphi_n(1)) - d_{\mathrm{H}}(\varphi_n(1), \mathrm{id}) \ge 1 - \tfrac{2}{n},$$

by (35). Hence σ_n is a sofic approximation. □

TableCover `tableGroup_no_model` `exists_finitelyPresented_obstruction`

Theorem 9.3 (Finite multiplication-table covers)**.** *Let G be a finitely generated group that is not sofic. Then there is an infinite finitely presented group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

Proof. Finite groups are sofic, so G is infinite. By Lemma 9.2 there are a finite set $F \subseteq G$ with $1 \in F$ and an $\varepsilon_0 > 0$ such that G admits no (F, ε_0)-model; enlarging F, we may assume F is symmetric and generates G. Define the finitely presented group

$$H = H_F = \big\langle\, x_g\ (g \in F \cup F{\cdot}F) \;\big|\; x_1 = 1,\quad x_g x_h = x_{gh}\ (g, h \in F)\,\big\rangle. \tag{36}$$

The assignments $x_g \mapsto g$ satisfy the relations in G, and F generates G, so there is a surjection

$$\pi : H \twoheadrightarrow G, \qquad \pi(x_g) = g. \tag{37}$$

In particular H is infinite, and $\pi(x_g) = g \ne h = \pi(x_h)$ shows $x_g \ne x_h$ in H for distinct $g, h \in F \cup F{\cdot}F$.

Suppose H were sofic, with sofic approximation $\sigma_n : H \to \mathrm{Sym}(Y_n)$. Define $\varphi_n : F \cup F\cdot F \to \mathrm{Sym}(Y_n)$ by $\varphi_n(g) = \sigma_n(x_g)$. For $g, h \in F$, the relation $x_g x_h = x_{gh}$ holds in H, so

$$d_{\mathrm{H}}(\varphi_n(g)\varphi_n(h), \varphi_n(gh)) = d_{\mathrm{H}}(\sigma_n(x_g)\sigma_n(x_h), \sigma_n(x_g x_h)) \to 0.$$

For distinct $g, h \in F$, the element $x_g x_h^{-1} \neq 1$ of H gives, as in the proof of Lemma 9.2,

$$d_{\mathrm{H}}(\varphi_n(g), \varphi_n(h)) \geq d_{\mathrm{H}}(\sigma_n(x_g x_h^{-1}), \mathrm{id}) - o(1) \to 1.$$

Hence for n large, φ_n is an (F, ε_0)-model of G—note that (34) and (35) refer only to the multiplication table of F inside G, which is faithfully copied by the generators of H. This contradicts the choice of (F, ε_0). Therefore H is not sofic. □

Theorem 9.4 (Kazhdan covers)**.** *Let G be a nonsofic group with property (T). Then there is an infinite finitely presented property-(T) group H that is not sofic, together with a surjection $H \twoheadrightarrow G$.*

ShalomFinitePresentation
`exists_presented_kazhdan_cover`
`exists_finitelyPresented_kazhdan_cover`
KazhdanCover
`exists_kazhdan_finitelyPresented_cover_of_not_isSofic`
Shalom's theorem is proved internally, not cited.

Proof. Property-(T) groups are finitely generated, and G is infinite as before. By Shalom's theorem [19], G is a quotient of a finitely presented property-(T) group: fix a surjection $\theta : P \twoheadrightarrow G$ with P finitely presented and Kazhdan. Choose (F, ε_0) as in the proof of Theorem 9.3, and for each $g \in F \cup F\cdot F$ choose a lift $\tilde{g} \in P$ with $\theta(\tilde{g}) = g$, taking $\tilde{1} = 1$. Define

$$H = P \,/\, \langle\!\langle\, \tilde{g}\,\tilde{h}\,\widetilde{gh}^{\,-1} \;:\; g, h \in F \,\rangle\!\rangle. \tag{38}$$

Then H is finitely presented (finitely many new relators over the finitely presented P) and has property (T) (a quotient of P). Each relator lies in $\ker\theta$, since $\theta(\tilde{g}\tilde{h}\widetilde{gh}^{\,-1}) = gh(gh)^{-1} = 1$; hence θ descends to a surjection $H \twoheadrightarrow G$, and H is infinite. Writing $y_g \in H$ for the image of $\tilde{g}$, we have $y_1 = 1$, $y_g y_h = y_{gh}$ for $g, h \in F$, and $y_g \neq y_h$ for distinct $g, h \in F$ (their images in G differ). If H were sofic, then $\varphi_n(g) = \sigma_n(y_g)$ would be an (F, ε_0)-model of G for large n, exactly as in the proof of Theorem 9.3, a contradiction. □

Proof of Theorem C. The first two assertions are Theorems 9.3 and 9.4. For the final assertion, apply Theorem 9.4 to the infinite finitely generated property-(T) nonsofic group $L^\times$ of Theorem B, over any fixed finite field; the resulting H surjects onto $L^\times$, and composing with the inverses of the isomorphisms $\Theta_{\mathcal{C}}$ of Theorem B(ii) exhibits a quotient of H isomorphic to each group listed there. □

Remark 9.5 (Effectivity)**.** The covers are effective in the logical sense: nonsoficity of G is equivalent, by Lemma 9.2, to the existence of a finite witness (F, ε_0), and any witness yields H by (36) or (38). The proof establishes the existence of a witness without computing one, and no claim is made that $L^\times$ itself is finitely presented.

10. Concluding remarks

Remark 10.1 (The trust surface of Theorem A)**.** Theorem A is independent of $\mathrm{GL}_r = \mathrm{EL}_r$ and of $K_1(L) = 0$: the memberships of u and z are explicit factorizations, the remaining memberships follow from the Whitehead identity and the perfectness of V, and the only analytic inequality proved in the paper is the co-area estimate of Lemma 2.7—the approximation-theoretic depth resides in the cited theorems of Kun and Kun–Thom. The argument concerns permutation models only; hyperlinearity of these groups, and the Connes embedding problem, remain open.

Remark 10.2 (Two compressors, one image)**.** The pair $Q = \{u, v\}$ resolves a genuine tension: generation of G requires elements that move Γ, while compression requires elements that map Γ into itself. The present realization divides the two roles. The compressor u seats $K = u\Gamma u^{-1}$; the involution $z = vu^{-1}$ generates the missing last row and column of EL_n (Proposition 6.4); and the two compressors share the single image K, so the criterion sees one coherent compression while z carries the generation. The criterion itself is indifferent to the size of Q; the algebra here simply supplies two natural compressors rather than one.

Remark 10.3 (The median is canonical up to scale)**.** The normalization $f = s/(s + m)$ of Section 4 pins at $1/2$, and the value $1/2$ is not a trace or a spectral constant: it is the fixed point of the exchange of source and target sizes, and it is a vertex-weighted median by construction. Median pinning is exactly the statement that a permutation compressing one expander decomposition into another must asymptotically preserve the size profile—the conservative core of the argument, with exact conservation (Lemma 3.2) converting a one-sided drift estimate into a two-sided one at no cost.

Remark 10.4 (What property (T) is used for)**.** Property (T) enters three times, in three different roles: (a) through Theorem 2.9 applied to Γ, producing the source decomposition that is compressed; (b) through Theorem 2.9 applied to G, producing the ambient decomposition carrying the co-area inequality—and by Remark 4.3 this use is replaceable by any componentwise ℓ^1 Poincaré decomposition; (c) through the Kazhdan factor hypothesis of Theorem 2.10. The local Theorem 4.1 quantifies over the first two uses and leaves only the third to the black box.

Remark 10.5 (Scope)**.** The Leavitt-corner theorem (Theorem 6.7) applies to an arbitrary nontrivial unital ring carrying a binary Leavitt family, whenever the two adjacent elementary groups are Kazhdan and the two matrices u, z are elementary; for finitely generated rings and $m \geq 3$ the Kazhdan hypothesis is automatic (Corollary 6.8). Finite fields are where all hypotheses are verified simultaneously (Theorem 8.8); characteristic two is where they are verified explicitly (Theorem 7.3); and the construction can already be carried out in rank two (Theorem 8.1).

Appendix A. An elementary and effective proof of $K_1(L_k(1,2)) = 0$

This appendix proves the vanishing of K_1 announced in Remark 8.5, in the Whitehead form in which Section 8 uses it, by elementary means: no localization sequence, no spectra, and no input beyond the two Leavitt relations and one classical fact about L recorded as Lemma A.3 below. The proof is an algorithm—a Gaussian elimination over the binary tree—and the complexity accounting of Remark 8.6 reads off from it. Every lemma of this appendix is formalized in the accompanying Lean development, and the margin notes record the correspondence; the appendix is written to be read and checked entirely on paper.

Throughout, k is a field and $L = L_k(1,2)$. Recall from Section 5 the word notation s_w, s_w^* (we write $t_w = s_w^*$ for the word w, so $t_w s_{w'} = \delta_{w,w'}$ for $|w| = |w'|$), the cylinder idempotents $p_w = s_w t_w$, and complete leaf sets. A *complete code of depth* $\leq r$ (resp. $\geq r$) is a complete leaf set all of whose words have length $\leq r$ (resp. $\geq r$).

A.1. The Whitehead class group.

Definition A.1. Let

DiagonalClassGroup
`stableUnits`
`ScalarReduction`

$$H \;=\; \left\{ u \in L^\times \,\middle|\, \operatorname{diag}(u,1) \in \mathrm{EL}_2(L) \right\}.$$

Since $\operatorname{diag}(uv,1) = \operatorname{diag}(u,1)\operatorname{diag}(v,1)$, H is a subgroup of $L^\times$. The main theorem of this appendix is:

Theorem A.2. *$H = L^\times$. Equivalently, for every $u \in L^\times$ and every $r \geq 2$, the matrix $\operatorname{diag}(u,1,\ldots,1)$ is a product of elementary matrices in $\mathrm{EL}_r(L)$.*

RefineLoopDischarge
`K1_trivial`
`narrowReduction_holds`
`scalarReduction_holds`
`stableUnits_eq_top_holds`

(The two forms are equivalent because $\mathrm{EL}_2 \hookrightarrow \mathrm{EL}_r$ in the first two coordinates.) The single external input is the following division property, a standard consequence of pure infiniteness and simplicity of L [1]; the accompanying formalization proves it from scratch, so a reader may take it either from the literature or from the machine-checked development.

Lemma A.3 (Strong division)**.** *For every nonzero $x \in L$ there are $p, q \in L$ with $pxq = 1$.*

LeavittSimplicity
`exists_mul_mul_eq_one`

The first consequence is a Gaussian elimination for invertible matrices, which we prove at every rank; it is the GE property of Theorem 8.2(a) in elementary form.

Lemma A.4 (Elementary diagonalization)**.** *Let A be a nontrivial unital ring in which every nonzero element x admits p, q with $pxq = 1$, and let $X \in \mathrm{GL}_r(A)$, $r \geq 2$. Then there are $E, F \in \mathrm{EL}_r(A)$ and $u \in A^\times$ with*

$$E\,X\,F = \operatorname{diag}(u,1,\ldots,1).$$

formalized in part
MatrixDiagonalization
`exists_elementary_mul_diag`
`binaryLeavitt_exists_`
`elementary_mul_diag`
Rank two is formalized; the higher-rank induction is manuscript-only.

Proof. First let $r = 2$ and write $X = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$. Assume first $a \neq 0$, and pick p, q with $paq = 1$. Three elementary operations plant a literal 1 in the corner: the column operation $C_2 \mathrel{+}= C_1 \cdot \rho$ with $\rho = q(1-pb)$ replaces b by $b' = a\rho + b$, which satisfies $pb' = paq(1-pb) + pb = 1$; the row operation $R_2 \mathrel{+}= wR_1$ with $w = (1-d')p$, where $d' = c\rho + d$, replaces d' by $wb' + d' = (1-d')pb' + d' = 1$; and with a 1 at position $(2,2)$, the operations $R_1 \mathrel{+}= (-b')R_2$ and $C_1 \mathrel{+}= C_2 \cdot (-c'')$ (where c'' is the current $(2,1)$ entry) clear the off-diagonal entries. The result is $\operatorname{diag}(v,1)$ with $v \in A^\times$, since the product of invertible matrices is invertible and a triangular-free diagonal matrix is invertible exactly when its entries are. If $a = 0$, then some entry of X is nonzero (as X is invertible and $A \neq 0$), and one elementary row or column addition moves a nonzero element into position $(1,1)$; this reduces to the previous case.

For $r > 2$, induct. Pick any nonzero entry of X and, by one row and one column addition, arrange $x_{11} \neq 0$. Run the four operations above in the coordinate pair $(1,r)$: they involve only rows and columns 1 and r, and they produce a literal 1 at position (r,r). Using that 1, clear the rest of row r and column r by elementary operations. The matrix is now $\operatorname{diag}(X',1)$ with $X' \in \mathrm{GL}_{r-1}(A)$, and the inductive hypothesis applies; for $r-1 = 1$ the surviving corner is the unit u. □

DiagonalClassGroup
`conj_mem_elementaryGroup_of_division`
`stableUnits_normal`

Lemma A.5. $\mathrm{EL}_2(L)$ *is normal in* $\mathrm{GL}_2(L)$*, and* H *is normal in* $L^\times$*.*

Proof. Conjugation by an elementary matrix preserves EL_2; conjugation by $\operatorname{diag}(u,v)$ sends $x_{12}(a)$ to $x_{12}(uav^{-1})$ and $x_{21}(a)$ to $x_{21}(vau^{-1})$, hence also preserves EL_2; and by Lemma A.4 every element of $\mathrm{GL}_2(L)$ is a product of elementary matrices and one matrix $\operatorname{diag}(u,1)$. For the second claim, if $u \in H$ and $g \in L^\times$, then $\operatorname{diag}(gug^{-1},1) = \operatorname{diag}(g,1)\operatorname{diag}(u,1)\operatorname{diag}(g,1)^{-1} \in \mathrm{EL}_2(L)$ by the first claim. □

By Lemma A.5 the quotient $L^\times/H$ is a group; we write $[u]$ for the class of u, so that Theorem A.2 says $[u] = 1$ for every unit. Three families of moves generate equalities in $L^\times/H$.

StableUnitsGenerators
`mem_stableUnits_of_val_unipotent`
IncomparableUnipotents
`incomparableUnit_mem`

Lemma A.6 (Unipotent moves)**.** *Let* $a, b \in L$ *with* $ba = 0$*. Then* $1+ab \in H$*. In particular, for prefix-incomparable words* v, w *and any* $x \in L$*, the* cylinder unipotent $1 + s_v x t_w$ *lies in* H*.*

Proof. From $ba = 0$, $(ab)^2 = a(ba)b = 0$, so $1+ab$ is a unit with inverse $1-ab$, and the four-term identity

$$\operatorname{diag}(1+ab,\ 1) = x_{12}(a)\, x_{21}(b)\, x_{12}(-a)\, x_{21}(-b)$$

holds by direct block computation: $x_{12}(a)x_{21}(b) = \left(\begin{smallmatrix} 1+ab & a \\ b & 1 \end{smallmatrix}\right)$, multiplying by $x_{12}(-a)$ clears the $(1,2)$ entry using $-(1+ab)a + a = -a(ba) = 0$, and $x_{21}(-b)$ clears the $(2,1)$ entry. For the particular case take $a = s_v x$, $b = t_w$: then $ba = t_w s_v\, x = 0$ since v, w are incomparable. □

WhiteheadFlip
`flipUnit`
`diagPair_flipUnit_inv_mem`
`mul_flipUnit_inv_mem_stableUnits`

Lemma A.7 (The flip)**.** *Let* $x, y \in L$ *and suppose* $1 + xy \in L^\times$*. Then* $1 + yx \in L^\times$*, with inverse* $1 - y(1+xy)^{-1}x$*, and*

$$[\,1+xy\,] = [\,1+yx\,] \qquad \text{in } L^\times/H.$$

Proof. The stated inverse is checked directly: $(1+yx)\big(1-y(1+xy)^{-1}x\big) = 1+yx-y(1+xy)(1+xy)^{-1}x = 1$, and likewise on the other side. The matrix $\mathrm{diag}(1+xy,\,(1+yx)^{-1})$ factors into four elementary matrices,

$$x_{12}(x)\,x_{21}(y)\,x_{12}\big(-(1+xy)^{-1}x\big)\,x_{21}\big(-(1+yx)y\big),$$

again by direct computation. Hence $\mathrm{diag}((1+xy)(1+yx)^{-1},1) = \mathrm{diag}(1+xy,(1+yx)^{-1})\cdot\mathrm{diag}\big((1+yx)^{-1},(1+yx)\big)^{-1}\cdot$ (a Whitehead diagonal, Lemma 5.5) lies in $\mathrm{EL}_2(L)$, so $(1+xy)(1+yx)^{-1}\in H$. ☐

Lemma A.8 (Corner insertion)**.** *For a binary word w and $u\in L^\times$, the corner insertion $\kappa_w(u) = s_w u t_w + (1-p_w)$ is a unit, and $[\kappa_w(u)] = [u]$. Consequently every central unit lies in H.* *LeavittDiagonalClass* `kappaUnit_mul_inv_mem_stableUnits` `central_mem_stableUnits` `stableUnits_eq_top`

Proof. The matrix $X_w = \begin{pmatrix} s_w & 1-p_w \\ 0 & t_w\end{pmatrix}$ is invertible, with inverse $\begin{pmatrix} t_w & 0 \\ 1-p_w & s_w\end{pmatrix}$ (using $t_w s_w = 1$ and $(1-p_w)s_w = 0$), and it intertwines:

$$X_w\;\mathrm{diag}(u,1)\,X_w^{-1} = \mathrm{diag}(\kappa_w(u),\,1).$$

Thus $\mathrm{diag}(\kappa_w(u)u^{-1},1) = [X_w,\mathrm{diag}(u,1)]$, a commutator of invertible matrices, one of which is elementary-diagonalizable; by Lemmas A.4 and 5.5 such a commutator lies in $\mathrm{EL}_2(L)$, so $\kappa_w(u)u^{-1}\in H$.

For a central unit c, centrality gives $s_0ct_0 + s_1ct_1 = c(s_0t_0+s_1t_1) = c$; on the other hand a direct computation using $t_1s_0 = 0$ shows

$$\kappa_0(c)\,\kappa_1(c) = s_0ct_0 + s_1ct_1.$$

Hence $[c] = [\kappa_0(c)][\kappa_1(c)] = [c]^2$ in $L^\times/H$, and $[c]=1$. ☐

A.2. Code moves. Complete leaf sets play the role of block structures. Two further families of units lie in H.

Lemma A.9 (Code scalars)**.** *Let $\mathcal{D} = (w_1,\dots,w_\ell)$ be a complete leaf set and $G\in\mathrm{GL}_\ell(k)$ an invertible scalar matrix. Then* *CodeScalarMoves* `codeScalar_unit_mem`

$$\Sigma_{\mathcal{D}}(G) \;=\; \sum_{i,j} s_{w_i}\,G_{ij}\,t_{w_j}$$

is a unit of L lying in H.

Proof. The assignment $G\mapsto\Sigma_{\mathcal{D}}(G)$ is unital and multiplicative by the leaf identities (Lemma 5.3), so $\Sigma_{\mathcal{D}}(G)$ is a unit with inverse $\Sigma_{\mathcal{D}}(G^{-1})$. Over the field k, G is a product of transvections and one diagonal matrix (Gaussian elimination). A transvection $I+\lambda E_{ij}$, $i\neq j$, transports to $1+\lambda\, s_{w_i}t_{w_j}$, a cylinder unipotent in H by Lemma A.6 (leaves of a complete leaf set are pairwise incomparable). A diagonal matrix $\mathrm{diag}(d_1,\dots,d_\ell)$ transports to $\prod_i \kappa_{w_i}(d_i)$ with central scalars $d_i\in k^\times$; each factor lies in H by Lemma A.8. ☐

CodeChangeUnits
`codeChange_mem_stableUnits`
CodeChangeGlue
`codeBijection_mem_stableUnits`

Lemma A.10 (Code changes). *Let $(\tau_1, \dots, \tau_\ell)$ and $(\sigma_1, \dots, \sigma_\ell)$ be complete leaf sets of the same size. Then the* code change

$$\Omega \;=\; \sum_{i=1}^{\ell} s_{\tau_i} t_{\sigma_i}$$

is a unit of L lying in H.

Proof. Ω is a unit with inverse $\sum_i s_{\sigma_i} t_{\tau_i}$, by the leaf identities. Note that the Ω are precisely the images U_g of Thompson's group V under the embedding (23), together with their compositions with index bijections; the lemma is a generation theorem for the Higman–Thompson groupoid. Induct on ℓ. For $\ell = 1$ both sets are the empty word and $\Omega = 1$. For $\ell \geq 2$, the source set, being the leaf set of a finite full binary tree, contains a sibling pair $\sigma_a = w0$, $\sigma_b = w1$. Composing Ω with at most two *transpositions* aligns the pairing so that the targets of σ_a, σ_b are again a sibling pair $v0, v1$ in the same order; here a transposition is a code change that swaps two leaves x, y of one complete leaf set and fixes the rest, i.e. $T(x,y) = s_x t_y + s_y t_x + (1 - p_x - p_y)$, and $T(x,y) \in H$ because it is $\Sigma_{\mathcal{D}}(P)$ for the corresponding scalar permutation matrix P on any complete leaf set $\mathcal{D}$ refining $\{x, y\}$—a code scalar (Lemma A.9). After alignment, the two sibling terms merge through the completeness relation, $s_{v0} t_{w0} + s_{v1} t_{w1} = s_v (s_0 t_0 + s_1 t_1) t_w = s_v t_w$, producing a code change between complete leaf sets of size $\ell - 1$; the inductive hypothesis and normality of H conclude. ☐

PencilReshape
`reshaped_pencil_mem_iff`
`exists_reshaped_pencil`
`pencilVal_window_mem`

Corollary A.11 (Reshaping). *Let (R_i), (C_j), (P_i), (Q_j) be complete leaf sets, with $|R| = |P|$ and $|C| = |Q|$, and let $E_{ij} \in L$. If*

$$u = \sum_{i,j} s_{R_i} E_{ij} \, t_{C_j} \quad \text{and} \quad v = \sum_{i,j} s_{P_i} E_{ij} \, t_{Q_j}$$

are both units, then $v = \Omega_1 u \, \Omega_2$ for code changes $\Omega_1 = \sum_i s_{P_i} t_{R_i}$ and $\Omega_2 = \sum_j s_{C_j} t_{Q_j}$, and hence $u \in H \iff v \in H$.

Proof. $\Omega_1 u \Omega_2 = \sum s_{P_i} (t_{R_i} s_{R_a}) E_{ab} (t_{C_b} s_{C_j}) t_{Q_j} = v$ by the leaf identities; apply Lemma A.10 and normality. ☐

A.3. The degree filtration and width reduction. The defining relations of L are homogeneous for $\deg s_i = +1$, $\deg t_i = -1$, so the free-algebra grading descends: $L = \bigoplus_{d \in \mathbb{Z}} L_d$ is $\mathbb{Z}$-graded, with L_d spanned by the monomials $s_\alpha t_\beta$, $|\alpha| - |\beta| = d$. Write

$$W[a,b] = \operatorname{span}_k \{ s_\alpha t_\beta \mid a \leq |\alpha| - |\beta| \leq b \},$$

and, for $x \in L$, write $x = \sum_d x^{(d)}$ for its graded components. Elements of $W[0,0]$ are called *balanced*; by the leaf identities, the balanced elements supported at depth $\leq n$ are exactly the matrices $\sum_{\gamma,\delta} G_{\gamma\delta} \, s_\gamma t_\delta$ over the full code $\{0,1\}^n$ with scalar G, i.e. the image of $M_{2^n}(k)$ under Σ-transport—so

Lemma A.9 says: *balanced units are exactly the transported invertible scalar matrices, and they lie in* H[1].

Lemma A.12 (Width reduction)**.** *Every* $u \in L^\times$ *satisfies* $[u] = [u']$ *for some unit* u' *with value in* $W[-1, 1]$ *(a* narrow *unit).*

LeavittWindowReduction
`exists_window_reduction`
LeavittGradingSpans
`exists_corner_move`
BinaryLeavittWindow
`exists_narrow_representative`

Proof. Every element of L lies in some window $W[-M, N]$. We show: if $N \geq 2$, then $[u] = [u'']$ for a unit u'' with value in $W[-M, \max(N-1, 1)]$; the mirror statement for $M \geq 2$ follows by the same argument applied to the anti-automorphism θ of Lemma A.13 below, and iterating both lands in $W[-1, 1]$.

The engine is the *corner move*: for any $v, w \in L$,

$$\Big[\,u\,\Big] = \Big[\,s_0(u + vw)t_0 + s_0 v t_1 + s_1 w t_0 + s_1 t_1\,\Big],$$

obtained by transporting the 2×2 identity $\left(\begin{smallmatrix} 1 & v \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} u & 0 \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ w & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} u+vw & v \\ w & 1 \end{smallmatrix}\right)$ along $\Theta_{\mathcal{C}_2}$ of Proposition 5.4: the images of the two unipotent factors are the cylinder-type unipotents $1+s_0 v t_1$ and $1+s_1 w t_0$, which lie in H by Lemma A.6, and the image of $\mathrm{diag}(u, 1)$ is $\kappa_0(u)$, congruent to u by Lemma A.8.

Let x be the value of u, with top component $x^{(N)}$, $N \geq 2$. Since $y\, t_0 s_0 = y$ for every y, the top component factors as $x^{(N)} = b\, s_0$ with $b = x^{(N)} t_0 \in W[N-1, N-1]$. Apply the corner move with $v = -b$, $w = s_0$: the new value is

$$s_0(x - x^{(N)})t_0 \;-\; s_0 b t_1 \;+\; s_1 s_0 t_0 \;+\; s_1 t_1.$$

The first summand lies in $W[-M, N-1]$ (conjugation by $s_0(\cdot)t_0$ preserves degree), the second has degree $N-1$, and the last two have degrees 1 and 0. Hence the new unit has value in $W[-M, \max(N-1, 1)]$, as claimed. ☐

A.4. Windows die. The next stage shows that units confined to a *one-sided* window already lie in H. This is where the analytic heart of the argument sits.

Lemma A.13 (Transpose)**.** *The assignment* $s_i \mapsto t_i$, $t_i \mapsto s_i$ *extends to an anti-automorphism* θ *of* L *with* $\theta^2 = \mathrm{id}$*; it maps* $W[a, b]$ *onto* $W[-b, -a]$*, and* $u \in H \iff \theta(u) \in H$*.*

ThetaStable
`thetaMatUnit_mem_elementaryGroup`
`thetaMatUnit_diagUnit`

Proof. The relations are symmetric under the assignment composed with order reversal, so θ exists; it negates degree. Applying θ entrywise to the transpose of matrices sends $x_{ij}(a)$ to $x_{ji}(\theta(a))$ and $\mathrm{diag}(u, 1)$ to $\mathrm{diag}(\theta(u), 1)$, so it preserves EL_2 and hence membership in H. ☐

Lemma A.14 (Pure tails are nilpotent)**.** *Let* $1 + \eta \in L^\times$ *with* $\eta \in W[1, 1]$*. Then* η *is nilpotent.*

PureTailNilpotency
`pure_tail_nilpotent`

---

[1]A balanced element that is a unit of L has invertible scalar matrix at its own depth—a singular matrix over a field is an annihilator of a nonzero matrix, hence a zero divisor—and therefore has balanced inverse.

Proof. Let $y = (1+\eta)^{-1}$, with graded components $y^{(d)}$ supported in a finite window. The equation $(1+\eta)y = 1$ reads degreewise: $y^{(d)} + \eta\, y^{(d-1)} = \delta_{d,0}$. From the bottom of the window upward, $y^{(d)} = 0$ for d below the window; then $y^{(d)} = -\eta y^{(d-1)}$ for $d < 0$ forces all negative components to vanish, $y^{(0)} = 1$, and $y^{(d)} = (-\eta)^d$ for $d > 0$. Since the window is finite, $(-\eta)^d = 0$ for large d. $\square$

NilpotentTailKill
`nilpotent_tail_mem_stableUnits`
RankNormalForm
`exists_rank_normal_form`

Lemma A.15 (Nilpotent tails die)**.** *Let $z \in L$ be balanced with $s_1 z$ nilpotent, and let $u \in L^\times$ have value $1 + s_1 z$. Then $u \in H$.*

Proof. Induct on the nilpotency index D of $s_1 z$; for $D = 1$, $z = t_1(s_1 z) = 0$ and $u = 1$. For $D \geq 2$, put $r = t_1(s_1 z)^{D-1} \in W[D-2, D-2]\,(\cdot)$; factoring $r = R\sigma$ with $R = rt_0^{D-2}$ balanced (as in Lemma A.12) and taking a balanced pseudo-inverse Ξ of R—balanced elements are scalar matrices along a full code, and every scalar matrix R has a matrix Ξ with $R\Xi R = R$—one obtains an idempotent-like element e with $e\,(s_1 z) = 0$ and $(s_1 z)^{D-1} e = (s_1 z)^{D-1}$. The *mover* $1 - s_1(ze)$ has square-zero tail, hence lies in H by Lemma A.6, and

$$(1 + s_1 z)(1 - s_1(ze)) = 1 + s_1(z - ze)$$

because $z\,e\,s_1(ze)$-cross terms vanish by $e(s_1 z) = 0$. The new tail satisfies $(s_1(z - ze))^{D-1} = (s_1 z)^{D-1}(1 - e) = 0$, so its index is at most $D - 1$, and the inductive hypothesis applies. $\square$

ZeroKOne
`balanced_component_isUnit`

Lemma A.16 (The keystone)**.** *Let $u = c + \zeta \in L^\times$ with c balanced and $\zeta \in W[1,1]$. Then $c \in L^\times$.*

Proof. Suppose not. Balanced elements at depth n are scalar matrices $M_{2^n}(k)$ along the full code; by the rank normal form over the field k there are balanced units g, h with $gch = e := \sum_{\gamma \in S} p_\gamma$, a cylinder idempotent over a subset S of the depth-n code, and c is invertible exactly when S is everything. So assume S proper, and let $f = 1 - e = \sum_{\gamma \in T} p_\gamma$ with $T \neq \emptyset$. Replace u by $w = guh = e + \zeta'$ (with $\zeta' = g\zeta h \in W[1,1]$); invertibility of c is not affected. Let $y = w^{-1}$, with graded components $y^{(d)}$, $d \in [\mathrm{lo}, \mathrm{hi}]$, and let $M = -\mathrm{lo}$.

The equation $wy = 1$ reads degreewise $e\, y^{(d)} + \zeta' y^{(d-1)} = \delta_{d,0}$. Multiplying the $d = 0$ equation by f on both sides ($fe = 0$, $f^2 = f$) gives $f = f\zeta'\, y^{(-1)} f$, and substituting the relation $y^{(d)} = fy^{(d)} - \zeta' y^{(d-1)}$ (valid for $d \leq -1$, from $ey^{(d)} = -\zeta' y^{(d-1)}$) repeatedly yields, after M steps (the components below the window vanish),

$$f \;=\; \sum_{j=0}^{M-1} \underbrace{f\zeta'(-\zeta')^j}_{\deg = j+1} \cdot \underbrace{f\, y^{(-1-j)} f}_{\deg = -1-j}. \tag{39}$$

Now count ranks at a deep uniform interface. For ℓ large, every factor in (39) is represented by a scalar matrix between the full codes of appropriate depths: an element of $W[d,d]$ supported at bounded depth is, for all large q, a $2^{q+d} \times 2^q$ scalar matrix (pad with the completeness relation). Representing the j-th summand at interface depths $\ell \to \ell - 1 - j \to \ell$, the middle factor passes through the matrix of f at depth $\ell - 1 - j$, whose rank is exactly $|T|\, 2^{\ell-1-j-n}$. Hence the matrix of the right-hand side of (39) at depth ℓ has rank at most $\sum_{j<M} |T|\, 2^{\ell-1-j-n} < |T|\, 2^{\ell-n}$—a geometric series short of its limit—while the matrix of f at depth ℓ has rank exactly $|T|\, 2^{\ell-n}$. This contradiction proves the lemma. □

Proposition A.17 (Windows die). *A unit of L whose value lies in $W[0, N]$ for some N, or in $W[-N, 0]$, lies in H.*

WindowNonnegReduction
`window_nonneg_mem_stableUnits`

Proof. By the transpose (Lemma A.13) it suffices to treat $W[0, N]$; induct on N.

WindowNonposReduction
`window_nonpos_mem_stableUnits`

Base, $N \le 1$. Let $u = c + \zeta$, c balanced, $\zeta \in W[1, 1]$. By the keystone (Lemma A.16) c is a balanced unit, so $c \in H$ with balanced inverse, and $u = c(1 + \eta)$ with $\eta = c^{-1}\zeta \in W[1, 1]$. By Lemma A.14 η is nilpotent. The corner insertion $\kappa_1(1 + \eta) = 1 + s_1(\eta t_1)$ is congruent to $1 + \eta$ (Lemma A.8), has balanced tail coefficient $z = \eta t_1$, and $s_1 z = s_1 \eta t_1$ inherits nilpotency from η (indeed $(s_1\eta t_1)^j = s_1 \eta^j t_1$); Lemma A.15 finishes.

WidthTwoReduction
`window_zero_one_mem_stableUnits`

Step, $N \ge 2$. Write the value as $1 + (a + \eta)$ with $a \in W[0, N - 1]$—subtracting 1 costs nothing since 1 is balanced—and η the top component, of degree N. Split $\eta = s_0 q_0 + s_1 q_1$ with $q_z = t_z \eta$ of degree $N - 1$. Perform the stable Whitehead block move along the four depth-two cylinders: with

$$X_z = s_{00}\, s_z\, t_{w_z}, \qquad Y_z = s_{w_z}\, q_z\, t_{00}, \qquad (w_0, w_1) = (01, 10),$$

set $m_2 = (1 - X_0)(1 - X_1)$ and $m_1 = (1 + Y_0)(1 + Y_1)$, products of cylinder unipotents in H (Lemma A.6), and let $\kappa = \kappa_{00}(u) = 1 + s_{00}(a + \eta)t_{00}$, congruent to u. In the product $m_2\, \kappa\, m_1$ all cross terms vanish by incomparability of the three cylinders $00, 01, 10$, except

$$X_0 Y_0 + X_1 Y_1 = s_{00}\, (s_0 q_0 + s_1 q_1)\, t_{00} = s_{00}\, \eta\, t_{00},$$

which cancels the top component inside κ. The value of $m_2 \kappa m_1$ is therefore

$$1 + \big(s_{00}\, a\, t_{00} \;-\; X_0 - X_1 \;+\; Y_0 + Y_1\big),$$

whose components have degrees in $[0, N - 1]$: the a-part keeps its degrees, $\deg X_z = 1$, and $\deg Y_z = N - 1$. By induction $m_2 \kappa m_1 \in H$, hence $u \in H$. □

A.5. The pencil and the elimination loop.

Lemma A.18 (Pencil form). *Let $x \in W[-1, 1]$. Then there are m and scalar matrices A_0, A_1, C, B_0, B_1, indexed by the full code $\mathcal{F} = \{0, 1\}^{m+1}$, with*

PencilForm
`exists_pencil_form`
RefineLoopDischarge
`pencilEntry_mem_window`
`codePair_expansion`

$$x = \sum_{i,j \in \mathcal{F}} s_i\, E_{ij}\, t_j, \qquad E_{ij} = A_{0,ij}\, t_0 + A_{1,ij}\, t_1 + C_{ij} \cdot 1 + B_{0,ij}\, s_0 + B_{1,ij}\, s_1.$$

Proof. Choose m with all monomials of x of depth $\le m$, and expand $x = \sum_{i,j} s_i (t_i x s_j) t_j$ along the full code of depth $m + 1$ (completeness relation on both sides). For a monomial $s_\alpha t_\beta$ of x and code words i, j of length

$m+1$: $t_i s_\alpha$ vanishes unless α is a prefix of i, in which case it is $t_{i'}$ with $|i'| = m+1-|\alpha| \geq 1$, and symmetrically $t_\beta s_j = s_{j'}$ with $|j'| = m+1-|\beta| \geq 1$; the product $t_{i'} s_{j'}$ collapses, by the prefix calculus, to 0, a scalar, or a single letter s_z or t_z, according to the degree $|\alpha|-|\beta| \in \{-1, 0, 1\}$. Hence each corner entry $t_i x s_j$ lies in the five-dimensional scalar pencil space displayed. □

We call a unit in this form a *pencil unit* over the code pair (rows, columns); the data is the quintuple of scalar matrices, and the *stack* is the vertical block matrix $\binom{B_0}{B_1}$.

StrictNegativePencil
`entry_window_negative_of_B_full`

Lemma A.19 (Strict negativity)**.** *Let u be a pencil unit over complete leaf sets (R_i), (C_j), and suppose the stack admits a scalar left inverse: matrices G_0, G_1 with $G_0 B_0 + G_1 B_1 = I$. Then there is N such that every corner entry of the inverse,*

$$y_{ji} = t_{C_j}\, u^{-1}\, s_{R_i},$$

lies in $W[-N, -1]$.

Proof. All y_{ji} lie in a common finite window; let d^* be the largest degree in which any y_{ji} has a nonzero component, and suppose $d^* \geq 0$. From $u\,u^{-1} = 1$, sandwiching and inserting the completeness relation for (C_j),

$$\sum_j E_{ij}\, y_{ji'} = \delta_{ii'} \qquad (\text{all } i, i').$$

Take the component of degree $d^*+1 \geq 1$ on both sides. On the left only the degree-$+1$ part of E can contribute against $y^{(d^*)}$—components of y above d^* vanish—so

$$\sum_j (B_{0,ij} s_0 + B_{1,ij} s_1)\, y^{(d^*)}_{ji'} = 0.$$

Multiplying on the left by t_0 and by t_1 separates the two shifts: $\sum_j B_{0,ij} y^{(d^*)}_{ji'} = 0$ and $\sum_j B_{1,ij} y^{(d^*)}_{ji'} = 0$; that is, the stack annihilates the matrix $Y = (y^{(d^*)}_{ji'})$ on the left. Applying the scalar left inverse, $Y = (G_0 B_0 + G_1 B_1) Y = 0$, contradicting the choice of d^*. Hence all components have degree ≤ -1. □

RefineStep
`refine_column`
`pencilEntry_mul_s`
GLVectorNormalization
`exists_isUnit_matrix_col`

Lemma A.20 (Refinement step)**.** *Let u be a pencil unit over (R, C) whose column j_0 is* stack-free*: $B_{0,ij_0} = B_{1,ij_0} = 0$ for all i. Then u is also a pencil unit over the code pair (R, C'), where C' replaces the word C_{j_0} by its two children $C_{j_0}0$, $C_{j_0}1$, with data given columnwise by*

$$(A_0, A_1, C, B_0, B_1)_{\cdot\, j_0 z} = (0,\ 0,\ A_{z,\cdot j_0},\ [z{=}0]\, C_{\cdot j_0},\ [z{=}1]\, C_{\cdot j_0}),$$

all other columns unchanged.

Proof. Insert $1 = s_0 t_0 + s_1 t_1$ before $t_{C_{j_0}}$: each term $s_{R_i} E_{ij_0} t_{C_{j_0}}$ becomes $\sum_z s_{R_i}\, (E_{ij_0} s_z)\, t_{C_{j_0} z}$, and for a stack-free entry the split identity

$$(a_0 t_0 + a_1 t_1 + c)\, s_z = a_z + c\, s_z$$

shows $E_{ij_0}s_z$ is again a pencil entry, with the A-data falling to the constant slot and the constant to the B-slot—the index shift of the Wiener–Hopf analogy. □

Lemma A.21 (Code supply)**.** *For $1 \le \kappa \le 2^r$ there is a complete leaf set of size κ and depth $\le r$; for $\kappa \ge 2^M$ there is one of size κ and depth $\ge M$.*

CodeShapeSupply
`exists_shallow_code`
`exists_deep_code`

Proof. Kraft counting. Shallow: start from the full code of depth $\lceil \log_2 \kappa \rceil \le r$ and merge sibling pairs until κ leaves remain (each merge reduces the count by one and cannot increase depth). Deep: start from the full code of depth M, of size $2^M \le \kappa$, and split leaves (each split increases the count by one and produces words of length $\ge M$). □

Theorem A.22 (The elimination loop)**.** *Every pencil unit over a row code of 2^r words lies in H.*

RefineLoopDischarge
`pencil_free_exit`
`pencil_full_exit`
`pencil_unit_mem_pow`

Proof. Let u be a pencil unit over (R, C), $|R| = 2^r$, $|C| = \kappa$. We induct on $2^{r+1} - \kappa$ (when this is ≤ 0, the first case below applies).

Free exit ($\kappa \ge 2^{r+1}$). By Lemma A.21 choose a shallow complete code P of size 2^r and depth $\le r$, and a deep complete code Q of size κ and depth $\ge r + 1$. Reshape u to the pair (P, Q) (Corollary A.11). Each pencil entry lies in $W[-1, 1]$, so each monomial of the reshaped value has degree at most $|P_i| + 1 - |Q_j| \le r + 1 - (r + 1) = 0$: the reshaped unit lies in $W[-N', 0]$ for some N', hence in H by Proposition A.17, hence $u \in H$.

Full stack. If the stack has a scalar left inverse, Lemma A.19 puts every corner entry y_{ji} of u^{-1} in $W[-N, -1]$. Expand $u^{-1} = \sum_{j,i} s_{C_j} y_{ji} t_{R_i}$ and reshape to a shallow column code of depth $\le r+1$ (possible: $\kappa \le 2^{r+1}$ in this branch, or the free exit applies) and a deep row code of depth $\ge r$. Monomial degrees are at most $(r + 1) + (-1) - r = 0$, so $u^{-1} \in H$ by Proposition A.17, and $u \in H$.

Refine. Otherwise the stack, having no scalar left inverse, has a nonzero kernel vector v (over the field k, a matrix with trivial kernel has a left inverse). Choose an invertible scalar matrix G whose j_0-th column is v (extend v to a basis). The code scalar $\Sigma_C(G)$ lies in H (Lemma A.9), and $u \cdot \Sigma_C(G)$ is a pencil unit over (R, C) with data multiplied columnwise by G; its column j_0 is stack-free because v kills both B_0 and B_1. Apply the refinement step (Lemma A.20): a pencil unit over (R, C') with $\kappa + 1$ columns. The counter $2^{r+1} - \kappa$ has dropped; the inductive hypothesis gives $u \cdot \Sigma_C(G) \in H$, hence $u \in H$. □

Proof of Theorem A.2. Let $u \in L^\times$. By width reduction (Lemma A.12), $[u] = [u']$ with u' narrow; by the pencil form (Lemma A.18), u' is a pencil unit over the full code pair of depth $m + 1$, a row code of 2^{m+1} words; by the elimination loop (Theorem A.22), $u' \in H$. Hence $u \in H$. □

A.6. Consequences and effectivity.

Corollary A.23. *For every field k and $L = L_k(1, 2)$: $K_1(L) = 0$; and $\mathrm{GL}_r(L) = \mathrm{EL}_r(L)$ for every $r \ge 2$.*

RefineLoopDischarge
`glTwo_eq_elementary_holds`
`glFour_eq_elementary_holds`

Proof. For $X \in \mathrm{GL}_r(L)$, Lemma A.4 writes $EXF = \mathrm{diag}(u, 1, \ldots, 1)$, and Theorem A.2 places the diagonal factor in $\mathrm{EL}_r(L)$; hence $X \in \mathrm{EL}_r(L)$. The same normal form computes K_1: every class of $\mathrm{GL}_r(L)/\mathrm{EL}_r(L)$-stabilized is represented by $\mathrm{diag}(u, 1, \ldots, 1)$, which is elementary. □

The proof of Theorem A.2 is an algorithm, and every step is effective Gaussian elimination over k: the width reduction processes one graded component per corner move; the pencil form is a change of basis at depth $m+1$; each loop pass computes a kernel vector or a left inverse of the stack; and both exits are reshapes along explicitly constructed codes followed by the window inductions, each step of which is again explicit. For a unit supported on monomials of depth at most d, all codes involved have at most 2^{d+2} words and the loop executes at most 2^{d+2} times, giving the complexity bound of Remark 8.6: the elementary factorization of $\mathrm{diag}(u, 1, \ldots, 1)$ is computed in time polynomial in 2^d and in the number of monomials of u.

Appendix B. The Lean formalization

Nothing in this appendix is used anywhere in the paper. The mathematics of Sections 1–10 is complete and self-contained as it stands, and a reader with no interest in machine verification may stop at Remark 10.5 without having skipped an argument. What follows records a second, independent check on the same mathematics, and says exactly how far that check reaches.

What the development is. The accompanying development

`https://github.com/SauersML/nonsofic_existence`

is a Lean 4 library (toolchain `v4.32.2`, on `mathlib4` at the same tag) of roughly 260 modules. It is not a transcription of this paper. It is an independent formal proof of the same headline theorem, and in several places it proves more than the paper claims: Kun's theorem (Theorem 2.9) and the Kun–Thom obstruction (Theorem 2.10) are cited here but *proved internally* there, and Proposition 5.12 is proved there for V itself, with no appeal to Higman's presentation. Where the two differ, the margin notes say so.

The public endpoints are

`nonsofic_groups_exist`, `universalLeavittEL4_not_isSofic`,
`exists_finitelyPresented_nonsofic_group`,

all in the module `MainResults`. Each is a closed theorem: it takes no hypothesis naming Kun, Kun–Thom, or Ershov–Jaikin-Zapirain, and no Nonempty premise standing in for a construction. The property-(T), compression, non-LEF, and expander-decomposition inputs are all built inside the library.

How to read the margin notes. Every numbered result of the paper—every definition, lemma, proposition, corollary, and theorem—carries a note in the margin. Remarks carry one only when they have formal content of their own; the concluding commentary does not.

A note in its plain form is a module name in italic, hyperlinked to its source file, followed in typewriter by the declarations in that module that carry the content. That is the whole claim: those declarations formalize this statement. A line underneath, when there is one, records a way in which the Lean statement is *stronger* than the one printed here—that Kun's theorem is proved rather than assumed, say.

Two phrases override the plain reading, and they appear only where they are true.

formalized in part:
Only part of the statement is formalized. The line below says which part is, and which part is not.
not formalized:
The statement is not formalized, and the line below says why. This occurs exactly once, at Lemma 2.8, which the Lean proof does not need because it replaces the residual-finiteness route by a direct finite obstruction.

The notes are decoration in the strict sense: setting `\leanlinksfalse` in the preamble removes every one of them, leaving the text of the paper untouched.

The trust surface of the formalization. A machine proof is worth exactly what its axioms and its statements are worth, so both are audited mechanically rather than asserted.

Axioms. The module `Audit` prints the axiom report of every public endpoint on an ordinary build, and `scripts/Audit.lean` independently walks the transitive axiom closure of the whole project namespace and fails on anything outside classical Lean. Every endpoint reduces to `propext`, `Classical.choice`, and `Quot.sound`—the three axioms of classical mathematics in Lean. There are no `sorry`s, and the library builds with `warningAsError=true`.

Statements. An axiom audit cannot tell you that the theorem proved is the theorem you wanted; only reading the statements can. The declarations named in the margins are therefore the ones to read. Two definitional points carry most of the weight, and both are discharged as theorems rather than left as caveats: `isKazhdanPair_iff_complex` shows the real-orthogonal form of property (T) used in the library agrees with the complex-unitary form, with the same control set and tolerance, and `IsKazhdanPair.liftUniverse` removes the universe restriction, so `HasKazhdanPropertyT` is the textbook property.

Reproduction. `lake exe cache get && lake build` checks the library and prints the axiom reports. The margin notes are themselves checked on every build of this document by `scripts/check_lean_refs.py`, which fails if any note names a module or a declaration that the Lean source does not contain; so the correspondence recorded in the margins cannot silently rot as either side changes.

How the library is laid out. The margin notes give the statement-by-statement correspondence. The coarse structure behind them is the following; each layer is proved before the one below it uses it.

Principal modules	**Role**
`Sofic`, `LEF`, `Kazhdan`, `ElementaryGroup`	Hamming models, local embeddings, real-orthogonal property (T), and EL_n: the vocabulary of Section 2.
`KunFiniteMarkov–KunDecomposition`	Turns a property-(T) sofic model into negligible-edit uniform expander components: Theorem 2.9, proved rather than cited.
`Refinement`, `Pinning`, `Selection`, `MedianNormalization`	The five finite lemmas of Section 3 and the median normalization of Section 4.4.
`CompressionSetup`, `Criterion`, `MatchingSelection`, `CriterionAssembly`	Matches core and ambient components, localizes the product action, and closes Theorem 4.1 and Corollary 4.2.
`KunThomFiniteMarkov–KunThomTheorem`	Extracts LEF from exact-product expansion: Theorem 2.10, proved rather than cited.
`FreeRoot*`, `A2*`, `FreeElementaryPropertyT`, `FiniteFieldElementaryPropertyT`	Property (T) at rank three for finite-type algebras over every finite field, by an explicit A_2 spectral-gap computation and a finite-stage Fourier transport: the input of Theorem 5.7.
`LeavittRankEquivalence`	The elementary rank equivalence of Proposition 5.8, at every pair of positive ranks.
`ShalomFinitePresentation`, `KazhdanCover`	Shalom's finite-presentation theorem, proved internally by an ultralimit argument, and the Kazhdan covers of Theorem 9.4.
`UniversalLeavitt`, `UniversalRankFour`, `RankFourCompressors`	The Leavitt relations, the two explicit compressors, and the corner witness: Sections 5–7.
`RefineLoopDischarge`	The two-exit elimination of Appendix A, giving $K_1(L) = 0$ in Whitehead form unconditionally.
`MainResults`, `TableCover`, `Audit`	Closes nonsoficity, builds the finitely presented cover (Theorem 9.3), and audits statements and axioms.

What is not formalized. Beyond Lemma 2.8, which is absent because the Lean proof does not need it, the gaps are those marked *formalized in part* in the margins: the arbitrary-finitely-generated-ring form of Theorem 5.7 (the rank-three finite-field cases the endpoints use are proved, in every characteristic); part (b) of Theorem 8.2, which remains a cited input; the higher-rank induction of Lemma A.4 and with it the $\mathrm{GL}_r = \mathrm{EL}_r$ collapse of Proposition 8.3, formalized at ranks two and four rather than all ranks; the essential-expander definition (Definition 2.6), which the library realizes through the matching-certificate repair rather than as a definition of its own; and the complexity bounds of Remark 8.6, whose underlying elimination is formalized.

Every one of the paper's external inputs is therefore cited here but proved there. Kun's theorem and the Kun–Thom obstruction are flagged in the margins at Theorems 2.9 and 2.10; Shalom's theorem is proved as `exists_finitelyPresented_kazhdan_cover`, so the Kazhdan half of Theorem C rests on no external citation in the formalization; property (T) for elementary groups is proved at rank three over every finite field, which is all the endpoints use; and the classical facts about V are bypassed by the two-relator obstruction of Remark 5.13. What remains genuinely imported is only the full generality of Theorem 5.7—arbitrary finitely generated rings—which no endpoint needs.

The vanishing $K_1(L) = 0$ deserves one precise sentence, because it is the place where the formalization first changed the standing of a result in the body of the paper: the elimination now written out as Appendix A was formalized first, as `K1_trivial`, and the appendix is its paper form; for the purposes of Theorem B the localization sequence of [2] is confirmation rather than dependency.

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